Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

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Abstract

[To be continued...]

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List of symbols and abbreviations

AF	Antiferromagnetic
BCS	Bardeen-Cooper-Schrieffer (theory)
β	Inverse temperature
$\Delta^{(\gamma)}$	γ-wave component of the gap function
DoF	Degree(s) of freedom
EHM	Extended Hubbard model
EV(s)	Expectation value(s)
FS	Fermi surface
HF	Hartree-Fock
HFP(s)	Hartree-Fock parameter(s)
HM	Hubbard model
L_ℓ	Number of lattice sites in direction ℓ
LRT	Linear Response Theory
MFT	Mean-Field Theory
NN(s)	Nearest neighbor(s)
RWC(s)	Relevant Wick's contraction(s)
RMP(s)	Renormalized model parameter(s)
SC	Superconductor
SDW	Spin-density wave
SSB	Spontaneous symmetry breaking
T_c	Critical temperature

Chapter 1

Anti-Ferromagnetic instability

This chapter is devoted to an in-depth discussion of the AF long-range ordering in the EHM by the means of MFT. In the context of the 2D HM, discussed in App. A, AF ordering establishes as an unconventional metallic phase which is only stable at half-filling (as is discussed in Sec. A.1.7). The situation for the EHM is similar: the present analysis is carried out based on the argument that, given the symmetry structure of the non-local NN interaction, the fundamental features of the commensurate AF phase of the HM are effectively preserved, while the parameters are redefined by the interaction itself. The two most relevant effects we discuss in this chapter are the "gap complexification" (Sec. [Missing]) and the "hopping renormalization" (Sec. [Missing]), the latter being a general feature of the EHM common to all phases preserving (or, at most, weakening without destroying) the translational invariance, discussed in Sec. **??**.

1.1 The AF phase and its symmetries

When discussing the AF phase on the EHM square lattice, we must specify the type of antiferromagnetism we are dealing with. SDW antiferromagnets on a square lattice can emerge in a variety of ways, the simplest one – described in this text – being the so-called "commensurate". By this we mean essentially a kind of long range spin density ordering varying on the length scale of two lattice sites. This type of AF structure is particularly interesting on a square lattice because the phase transition is simply described in reciprocal space by the order parameter

$$\hat{\psi}_{\mathbf{k}\sigma} \equiv \hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}$$

going from a null EV in the normal phase to a non-zero value. This phenomenon occurs at the so called "AF vector"

$$\boldsymbol{\pi} \equiv (\pi, \pi)$$

Large portions of a square lattice Fermi surface (FS) can turn out to be connected precisely by an AF vector: this implies that, if the hamiltonian makes available a scattering channel at this wavevector, electrons can jump from one side to the other, making said EV non-zero.

In principle, the AF phase is physically established by breaking the SU(2) spin rotation symmetry down to the $\mathrm{U}^z(1)$ projected spin rotation symmetry; however, it is to be noted that – even if the concept of SSB is well defined only for the $\mathrm{SU}(2) \to \mathrm{U}^z(1)$ symmetry reduction – the hamiltonian in thermodynamic limit remains $\mathbb{Z}_2$ invariant with respect to spin inversion, $\sigma \leftrightarrow \overline{\sigma}$. Identical considerations as in Sec. **??** hold This: all physical quantities are spin-independent. As for the normal phase, we will use this considerations at the end of the calculations.

1.1.1 Antiferromagnetism in the HM

In this short section we recall the principal feature of the AF phase established in the conventional HM, extensively discussed in Sec. A.1. Before starting, let us define as in Eqns. (A.8) and (A.7) the AF and density fluctuations operators,

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} \tag{1.1}$$

$$\hat{\psi}_{\mathbf{k}\sigma} \equiv \hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \tag{1.2}$$

EV	Meaning	Equation
$\langle \hat{s}^x_{\mathbf{k}\sigma} \rangle$	$\left\langle \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^\dagger_{\mathbf{k}\sigma} \right\rangle$	(A.18)
$i \langle \hat{s}^y_{\mathbf{k}\sigma} \rangle$	$\left\langle \hat{\psi}_{\mathbf{k}\sigma} - \hat{\psi}^\dagger_{\mathbf{k}\sigma} \right\rangle$	(A.19)
$\langle \hat{s}^z_{\mathbf{k}\sigma} \rangle$	$\langle \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle$	(A.20)

Table 1.1 | List of the various expectation values (EVs) of each pseudospin component, their meaning in terms of fermionic operators and the reference equations from the pure HM.

which represent the two essential fermionic quadratic operators of the analytic MFT derivation. Within the pure HM, the AF phase is specified by the Ansatz (A.2) which is explicitly breaking translational invariance in each spin sector, while preserving $\mathrm{U}^z(1)$ and $\mathrm{U}^c(1)$ symmetries. As reported in Tab. **??**, the only RWC comes in Hartree channel, since the local two-body interaction couples operators at opposite spin index. The hamiltonian $\hat{H}$ is reduced thereby to the form of Eq. (A.3)

$$\hat{H}_t + \hat{H}_U \overset{\text{MFT}}{\simeq} -t \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_\sigma \left[\hat{c}^\dagger_{\mathbf{r}\sigma} \hat{c}_{\mathbf{r}'\sigma} + \text{h.c.} \right] + nU \sum_{\mathbf{r}} \left[\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow} \right] - mU \sum_{\mathbf{r}} (-1)^{x+y} \left[\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow} \right] - E^{(\text{AF})}_{\text{H}/U}$$

with $E^{(\text{AF})}_{\text{H}/U}$ the contraction energy resulting from double counting terms, given explicitly in Eq. (A.4). In reciprocal space, the hamiltonian decomposes as in Eq. (A.9),

$$\hat{H}_t + \hat{H}_U \overset{\text{MFT}}{\simeq} \sum_{\mathbf{k}\in\text{MBZ}} \sum_\sigma \hat{\Psi}^\dagger_{\mathbf{k}\sigma} h_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} - E^{(\text{AF})}_{\text{H}/U} \quad \text{being} \quad h_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \epsilon_{\mathbf{k}\sigma} & -\Delta_{\mathbf{k}\sigma} \\ -\Delta^*_{\mathbf{k}\sigma} & -\epsilon_{\mathbf{k}\sigma} \end{bmatrix}$$

and $\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU$. Nambu spinorial formulation is used,

$$\hat{\Psi}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{bmatrix}$$

and the free electrons energy is simply the tight binding energy

$$\epsilon_{\mathbf{k}\sigma} = -2t \left(\cos k_x + \cos k_y \right)$$

which is spin-invariant. The MFT description of the model reduces to a gas of free "γ-fermions", described by the Nambu spinor of Eq. (**??**),

$$\hat{\Gamma}_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} = \begin{bmatrix} \hat{\gamma}^{(+)}_{\mathbf{k}\sigma} \\ \hat{\gamma}^{(-)}_{\mathbf{k}\sigma} \end{bmatrix}$$

where

$$W_{\mathbf{k}\sigma} = \begin{bmatrix} \cos\theta_{\mathbf{k}\sigma} & -\sin\theta_{\mathbf{k}\sigma} \\ \sin\theta_{\mathbf{k}\sigma} & \cos\theta_{\mathbf{k}\sigma} \end{bmatrix} \quad \text{and} \quad \sin 2\theta_{\mathbf{k}\sigma} \equiv \frac{\Delta_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}}$$

These fermions populate the two bands $\pm E_{\mathbf{k}\sigma} = \sqrt{\epsilon^2_{\mathbf{k}\sigma} + |\Delta_{\mathbf{k}\sigma}|^2}$. The entire system is mapped onto an ensemble of pseudo-spins, each subject to a pseudo-field, as in Fig. 1.1. To diagonalize the system essentially means to anti-align each pseudo-spin with the z axis. Each pseudospin component expectation value (EV) is described by a simple equation and has a precise meaning, listed in Tab. 1.1.

1.1.2 Extension to the EHM

Consider now the non-local interaction $\hat{H}_V$: since only translational invariance is partially broken in the AF phase, the only RWCs within the MFT decomposition are Hartree terms and the same-spin

Fock term. The discussion is thus essentially the same as for the normal phase of Chap. **??**, with the only relevant difference being the Ansatz we use in Eq. (**??**), substituted by the commensurate oscillatory Ansatz of Eq. (A.2). As for the normal phase, by design the MFT solution to the model for the given phase is described by a redefinition of the various physical quantities and mathematical objects,

$$\epsilon_{\mathbf{k}\sigma} \to \tilde{\epsilon}_{\mathbf{k}\sigma} \qquad E_{\mathbf{k}\sigma} \to \tilde{E}_{\mathbf{k}\sigma} \qquad \Delta_{\mathbf{k}\sigma} \to \tilde{\Delta}_{\mathbf{k}\sigma} \qquad \mu \to \tilde{\mu}$$

The band energies renormalization is simply

$$\tilde{E}_{\mathbf{k}\sigma} \equiv \sqrt{\tilde{\epsilon}^2_{\mathbf{k}\sigma} + |\tilde{\Delta}_{\mathbf{k}\sigma}|^2}$$

and the following relations hold (the *tilde* sign indicates the renormalized quantity):

$$\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \tau^x \hat{\Psi}_{\mathbf{k}\sigma} \rangle = -\sin(2\tilde{\theta}_{\mathbf{k}\sigma}) \sin(2\tilde{\zeta}_{\mathbf{k}\sigma}) \langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \tag{1.3}$$

$$\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \tau^y \hat{\Psi}_{\mathbf{k}\sigma} \rangle = -\sin(2\tilde{\theta}_{\mathbf{k}\sigma}) \cos(2\tilde{\zeta}_{\mathbf{k}\sigma}) \langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \tag{1.4}$$

$$\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Psi}_{\mathbf{k}\sigma} \rangle = \cos(2\tilde{\theta}_{\mathbf{k}\sigma}) \langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \tag{1.5}$$

equivalent to the ones for the pure HM, Eqns. (A.18), (A.19) and (A.20). Moving to the the tilted frame, the z component of the pseudospin is just a thermal superposition of the "up" and "down" states

$$\langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle = \frac{1}{2} \left[f\left(\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) - f\left(-\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) \right] \tag{1.6}$$

Let us define the (renormalized) thermal factors as in Eq. (A.17),

$$\tilde{\mathrm{Th}}^{(\pm)}_{\mathbf{k}\sigma}[\beta, \tilde{\mu}] \equiv f\left(\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) \pm f\left(-\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) \tag{1.7}$$

The angles satisfy:

$$\sin(2\tilde{\xi}_{\mathbf{k}\sigma}) = \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{|\tilde{\Delta}_{\mathbf{k}\sigma}|} \quad \cos(2\tilde{\xi}_{\mathbf{k}\sigma}) = \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{|\tilde{\Delta}_{\mathbf{k}\sigma}|} \quad \sin(2\tilde{\theta}_{\mathbf{k}\sigma}) = \frac{|\tilde{\Delta}_{\mathbf{k}\sigma}|}{\tilde{E}_{\mathbf{k}\sigma}} \quad \cos(2\tilde{\theta}_{\mathbf{k}\sigma}) = \frac{\tilde{\epsilon}_{\mathbf{k}\sigma}}{\tilde{E}_{\mathbf{k}\sigma}} \tag{1.8}$$

The physical behaviour is the same as for the pure HM. In terms of MFT discussion, the AF phase is similar enough to the normal phase of Chap. **??**: the RWCs are the same of Sec. **??**, listed in Tabs. **??** and **??**. All three Hartree contractions, responsible in the normal phase of the chemical potential shift, and the s.s. Fock contraction, that originated the hopping renormalization. Let us start by the Hartree terms: due to the "staggered" nature of the commensurate AF phase, their role is here more complex. Then we will move to the Fock term, discussing the MFT approximations effect on bands. Next sections are basically an extension of the calculations of Chap. **??**.

1.2 Non-local Hartree effects: chemical potential renormalization

As in Eq. (**??**), the same-spin and opposite-spin non-local Hartree terms are

$$\hat{H}_V \simeq \overbrace{-V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{n}_{i\sigma} \rangle \hat{n}_{j\sigma} + \hat{n}_{i\sigma} \langle \hat{n}_{j\sigma} \rangle - \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\sigma} \rangle \right]}^{\text{s.s.}}$$
$$\underbrace{-V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{n}_{i\sigma} \rangle \hat{n}_{j\bar{\sigma}} + \hat{n}_{i\sigma} \langle \hat{n}_{j\bar{\sigma}} \rangle - \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\bar{\sigma}} \rangle \right]}_{\text{o.s.}} + (\text{all the rest})$$

Let $i \to \mathbf{r} = (x, y)$ and $j \to \mathbf{r}' = (x', y')$. The key difference in the present derivation, with respect to the normal phase, is the single-site density Ansatz: instead of Eq. (**??**), we use Eq. (A.2), which describes an oscillatory density for each spin sector,

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = n - (-1)^{x+y+\delta_{\sigma=\uparrow}} m$$

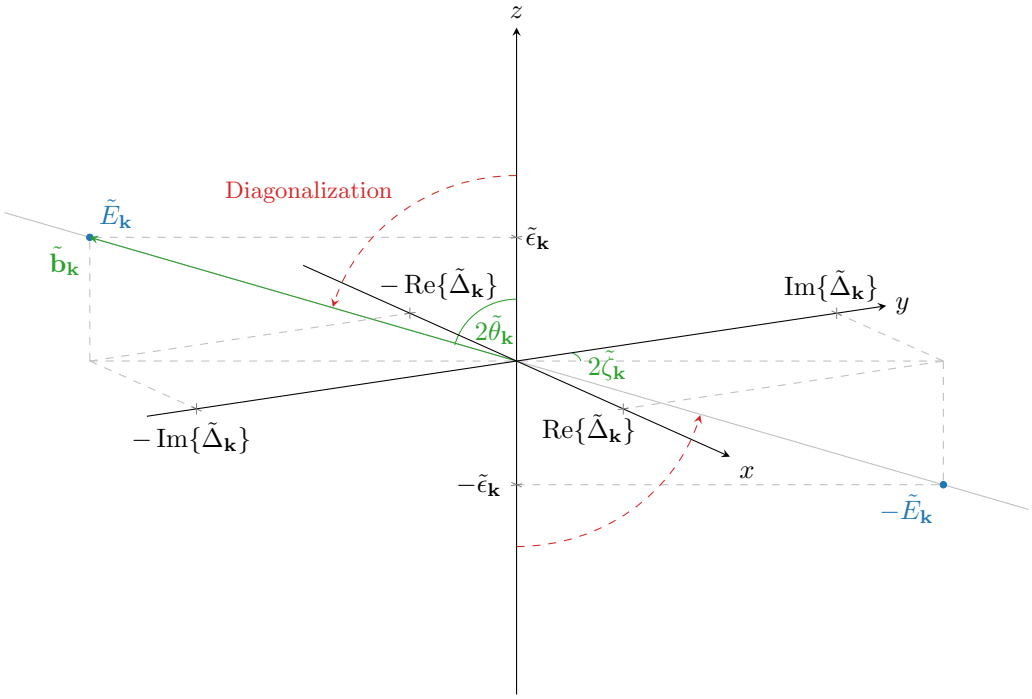


Figure 1.1 Sketch of the diagonalization of the pseudo-spin problem. The dashed line represents the diagonalization given by the z-axis alignment with the field direction. In the top-right side of the picture are listed the expectation values of the original Nambu spinor. For simplicity everywhere the index $\sigma = \uparrow$ has been omitted.

This Ansatz is composed of two parts: the offset n, and the oscillatory part m. The algebraic derivation relative to the n part is, of course, the same as for the normal phase. We now treat separately the "operatorial" part of the braced terms of Eq. (**??**),

$$-V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\langle \hat{n}_{\mathbf{r}\sigma} \rangle \hat{n}_{\mathbf{r}'\sigma} + \hat{n}_{\mathbf{r}\sigma} \langle \hat{n}_{\mathbf{r}'\sigma} \rangle] - V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\langle \hat{n}_{\mathbf{r}\sigma} \rangle \hat{n}_{\mathbf{r}'\overline{\sigma}} + \hat{n}_{\mathbf{r}\sigma} \langle \hat{n}_{\mathbf{r}'\overline{\sigma}} \rangle] \tag{1.9}$$

as well as the its scalar counterpart

$$V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\sigma} \rangle + V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\overline{\sigma}} \rangle \tag{1.10}$$

which accounts for the energy shift to be accounted when performing Wick's contractions.

1.2.1 Hartree channel in the AF phase: analytical derivation

Let us start from the operator of Expr. (1.9). By inserting the Ansatz of Eq. (A.2), we get

$$\begin{aligned}
&\overbrace{-nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}'\sigma} + \hat{n}_{\mathbf{r}\sigma}] + mV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} (-1)^{\delta_{\sigma=\uparrow}} \left[(-1)^{x+y} \hat{n}_{\mathbf{r}'\sigma} + (-1)^{x'+y'} \hat{n}_{\mathbf{r}\sigma} \right]}^{\text{s.s.}} \\
&\underbrace{-nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}'\overline{\sigma}} + \hat{n}_{\mathbf{r}\sigma}] + mV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \left[(-1)^{x+y+\delta_{\sigma=\uparrow}} \hat{n}_{\mathbf{r}'\overline{\sigma}} + (-1)^{x'+y'+\delta_{\overline{\sigma}=\uparrow}} \hat{n}_{\mathbf{r}\sigma} \right]}_{\text{o.s.}}
\end{aligned}$$

For a square lattice, if $\mathbf{r} = (x, y)$ and $\mathbf{r}' = (x', y')$ are NNs, evidently

$$(-1)^{x'+y'} = (-1)^{x+y+1}$$

Moreover,

$$(-1)^{\delta_{\overline{\sigma}=\uparrow}} = (-1)^{\delta_{\sigma=\uparrow}+1}$$

We obtain

$$\overbrace{-nV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\sigma}]}^{\text{s.s. (density)}}-\overbrace{mV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(-1)^{x+y+\delta_{\sigma=\uparrow}}[\hat{n}_{\mathbf{r}\sigma}-\hat{n}_{\mathbf{r}'\sigma}]}^{\text{s.s. (magnetization)}}$$
$$\underbrace{-nV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\overline{\sigma}}]}_{\text{o.s. (density)}}+\underbrace{mV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(-1)^{x+y+\delta_{\sigma=\uparrow}}[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\overline{\sigma}}]}_{\text{o.s. (magnetization)}} \tag{1.11}$$

In the above expressions the various contribution have been separated in "density" contributions and "magnetization" contributions. Let us deal with these separately.

Density terms. Consider the s.s. and o.s. (density) terms of Expr. (1.11),

$$-nV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\sigma}]-nV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\overline{\sigma}}] \tag{1.12}$$

The discussion is identical to the one of Sec. **??** for the normal phase, with the renormalization of μ being the same of Eq. (**??**).

$$\tilde{\mu} \equiv \mu + n(2zV - U) \tag{1.13}$$

Within this equation we are also already implicitly considering the μ shift effect due to $\hat{H}_U$.

Magnetization terms. The s.s. and o.s. (magnetization) terms of Expr. (1.11) perfectly cancel out:

$$mV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(-1)^{x+y+\delta_{\sigma=\uparrow}}[(\hat{n}_{\mathbf{r}\sigma}-\hat{n}_{\mathbf{r}'\sigma})-(\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\overline{\sigma}})]=0$$

thus the only net effect of the Hartree RWCs affects μ. This is somewhat to be expected: in the context of the HM, the gap function emerges in its purely s-wave structure from this precise computation on Hartree terms. However, the $\hat{H}_V$ non-local interaction cannot provide a pure s-wave potential component. Effects on the gap function are to be observed somewhere else.

Let us now move to the last part of the Hartree effects on the AF phase originated by $\hat{H}_V$, which is, the energy shift due to double counting terms originated by MFT decomposition of the quartic interactions $\hat{H}_U$ and $\hat{H}_V$.

1.2.2 Hartree contraction energy

Consider the scalar Expr. (1.10): it's the sum of two parts, the first coming from s.s. sector, the second from o.s. sector.

s.s. sector double counting energy. For the s.s. part, explicitating the two spin sectors, we get

$$V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\langle\hat{n}_{\mathbf{r}\sigma}\rangle\langle\hat{n}_{\mathbf{r}'\sigma}\rangle = \overbrace{V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\Big[(n+(-1)^{x+y}m)(n+(-1)^{x'+y'}m)\Big]}^{\sigma=\uparrow}$$
$$\underbrace{+V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\Big[(n-(-1)^{x+y}m)(n-(-1)^{x'+y'}m)\Big]}_{\sigma=\downarrow}$$

Evidently the mixed terms (those of order $\mathcal{O}(n)\times\mathcal{O}(m)$) cancel out, and considering that the sign factor is necessarily alternating on neighboiring site, the remainder is just

$$V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\sigma}\rangle = V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}2(n^2-m^2)$$

o.s. sector double counting energy. Proceeding identically on the o.s. sector, we get

$$V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\langle \hat{n}_{\mathbf{r}\sigma}\rangle\langle \hat{n}_{\mathbf{r}'\overline{\sigma}}\rangle = \overbrace{V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\Big[(n+(-1)^{x+y}m)(n-(-1)^{x'+y'}m)\Big]}^{\sigma=\uparrow} + \underbrace{V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\Big[(n-(-1)^{x+y}m)(n+(-1)^{x'+y'}m)\Big]}_{\sigma=\downarrow}$$

which gives immediately

$$V\sum_{\langle ij\rangle}\sum_{\sigma}\langle \hat{n}_{i\sigma}\rangle\langle \hat{n}_{j\overline{\sigma}}\rangle = V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}2(n^2+m^2)$$

Hence, the total "contraction energy", given by the sum of both contributions, is identical to the normal one of Eq. (**??**) indeed:

$$E^{(\mathrm{AF})}_{\mathrm{H}/V} = -2V\sum_{\langle ij\rangle}\sum_{\sigma}n^2 = 4n^2V\times\frac{z}{2}L_xL_y \tag{1.14}$$

the minus sign coming from the fact that we are subtracting this energy shift. This implies that, when comparing normal and AF phase, we can rule out this energy contribution, being equal for both free energy densities.

1.3 Fock effects: gap and bands renormalization

We explored thoroughly the Hartree contractions effects on the MFT hamiltonian. Let us now move to the effects originated by Fock contractions. We can use the derivation of Sec. **??** up to Eq. (**??**),

$$\begin{aligned} V\sum_{\langle ij\rangle}\sum_{\sigma}&\Big[\langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}+\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\langle \hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle-\langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\langle \hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle\Big] \\ &\overset{\mathcal{F}}{=}\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}\left[\cos(2k_x)+\cos(2k_y)\right]\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}\sigma}-E^{(\mathrm{AF})}_{\mathrm{F}/V} \end{aligned} \tag{1.15}$$

where $E^{(\mathrm{AF})}_{\mathrm{F}/V}$ is defined as in Eq. (**??**), with the expectation value being computed on the AF (thermal) ground-state. As before, we discuss the operatorial part and the scalar part separately.

1.3.1 Fock channel in the AF phase: analytical derivation

The commensurate AF phase is essentially described by two EVs being in principle non-zero,

$$\big|\langle \hat{n}_{\mathbf{k}\sigma}\rangle\big|\geq 0 \qquad \big|\langle \hat{\psi}_{\mathbf{k}\sigma}\rangle\big|\geq 0$$

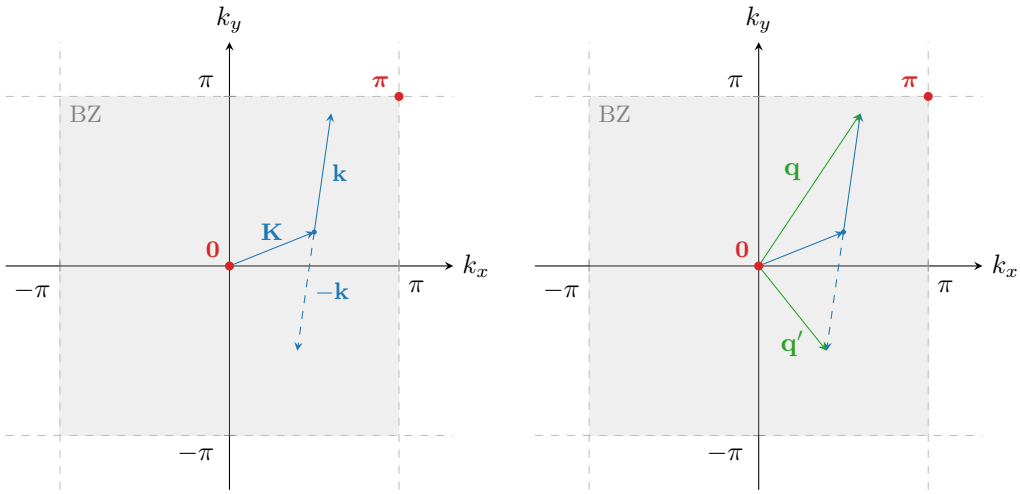
The AF ground state is a degenerate Fermi gas of $\gamma^{(\pm)}$ quasiparticles, and for null temperature acquires the simple form of Eq. (A.28). Going back to Eq. (1.15), this implies only two contributions are non-zero:

$$\mathbf{k}=-\mathbf{k}' \quad \text{or} \quad \mathbf{k}+\boldsymbol{\pi}=-\mathbf{k}'$$

The first case gives us the normal fermionic EV, $\langle \hat{n}_{\mathbf{k}\sigma}\rangle$, the second the staggered AF order parameter, $\langle \hat{\psi}_{\mathbf{k}\sigma}\rangle$. Then Eq. (1.15) is reduced to:

$$\begin{aligned}(\cdots)\overset{\mathcal{F}}{=}\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}&\left[\cos(2k_x)+\cos(2k_y)\right] \\ &\Big[\underbrace{\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}\sigma}}_{\text{Diagonal terms}}-\underbrace{\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}}_{\text{Off-diagonal terms}}\Big]\end{aligned} \tag{1.16}$$

Now, the above equation presents *diagonal* and *off-diagonal* terms.



a | Generic vectors. b | Change of variables.

Figure 1.2 | Representation of the vectors involved in the diagonal terms of Eq. (**??**). In Fig. 1.2a generic vectors are considered, cycling over all values of $\mathbf{K}, \mathbf{k} \in$ BZ. In Fig. 1.2b is depicted the variables change to the new vectors $\mathbf{q}, \mathbf{q}'$.

Diagonal terms. The diagonal terms of Eq. (1.16) are simple density interactions with the mean density field. Consider Fig. 1.2a: density at vector $\mathbf{q} \equiv \mathbf{K} + \mathbf{k}$ interacts with the mean density at vector $\mathbf{q}' \equiv \mathbf{K} - \mathbf{k}$. These variables are depicted in Fig. 1.2b. Apply in the diagonal part of Eq. (1.16) this variables change (which is the same of Eq. (**??**))

$$\begin{aligned}
\frac{2V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma} \\
&= \frac{2V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_{\sigma} [\cos(\delta q_x) + \cos(\delta q_y)] \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} \\
&\overset{*}{=} \sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{n}_{\mathbf{q}\sigma} \rangle \right] \times V \sum_{\mathbf{q}'} \varphi^{(\gamma)*}_{\mathbf{q}'} \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma}
\end{aligned} \tag{1.17}$$

where the sign $\overset{*}{=}$ indicates that we have substituted the decomposition of Eq. (**??**). Define as in Eq. (**??**) the EV

$$u_{\mathbf{q}\sigma} \equiv \langle \hat{n}_{\mathbf{q}\sigma} \rangle \tag{1.18}$$

The expression in square brackets of Eq. (1.17) is just its γ component, and is expanded in

$$\begin{aligned}
u^{(\gamma)}_{\sigma} &= \frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{n}_{\mathbf{q}\sigma} \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \left[\varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{n}_{\mathbf{q}\sigma} \rangle + \varphi^{(\gamma)}_{\mathbf{q}+\boldsymbol{\pi}} \langle \hat{n}_{\mathbf{q}+\boldsymbol{\pi}\sigma} \rangle \right] \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{q}} \left[\langle \hat{n}_{\mathbf{q}\sigma} \rangle - \langle \hat{n}_{\mathbf{q}+\boldsymbol{\pi}\sigma} \rangle \right] \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{\Psi}^\dagger_{\mathbf{q}\sigma} \tau^z \hat{\Psi}_{\mathbf{q}\sigma} \rangle \\
&= \frac{1}{2L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{q}} \frac{\tilde{\epsilon}_{\mathbf{q}\sigma}}{\tilde{E}_{\mathbf{q}\sigma}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta, \tilde{\mu}]
\end{aligned} \tag{1.19}$$

where, in the last passage, Eqns. (1.5), (1.6) and (1.7) have been used. Now, the AF phase does not break inversion symmetry: this implies that, whatever the shape of the renormalized bands, it

must be

$$\tilde{\epsilon}_{\mathbf{q}\sigma} = \tilde{\epsilon}_{-\mathbf{q}\sigma} \quad \Longrightarrow \quad u_\sigma^{(p_\ell)} = 0 \quad \text{with} \quad \ell \in \{x, y\}$$

The renormalized bands need to be symmetric functions of the moment. Hence for $\gamma \in \{p_x, p_y\}$ the above sum vanishes. Thus, Eq. (**??**) becomes

$$\frac{2V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_\sigma \left[\cos(2k_x) + \cos(2k_y)\right] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}$$
$$= V \sum_{\mathbf{q},\sigma} \left[u_\sigma^{(s^*)} (\cos q_x + \cos q_y) + u_\sigma^{(d)} (\cos q_x - \cos q_y) \right] \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma}$$

These two factors, together, define the full bands renormalization

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv \epsilon_{\mathbf{k}\sigma} + u_\sigma^{(s^*)} V (\cos q_x + \cos q_y) + u_\sigma^{(d)} V (\cos q_x - \cos q_y) \tag{1.20}$$

as was for the normal phase in Eq. (**??**). In Sec. **??** we discussed the appearance of anisotropic hopping with d-wave extension of the bands: here the theory predicts something identical. These RMPs are, as before, controlled linearly by V and in principle can lead to relevant changes in the kinetic sector of the hamiltonian.

Through this decomposition we have identified two HFPs and their self-consistency equations,

$$u_\sigma^{(s^*)} = \frac{1}{2L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} (\cos q_x + \cos q_y) \frac{\tilde{\epsilon}_{\mathbf{q}\sigma}}{\tilde{E}_{\mathbf{q}\sigma}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta, \tilde{\mu}] \tag{1.21}$$

$$u_\sigma^{(d)} = \frac{1}{2L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} (\cos q_x - \cos q_y) \frac{\tilde{\epsilon}_{\mathbf{q}\sigma}}{\tilde{E}_{\mathbf{q}\sigma}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta, \tilde{\mu}] \tag{1.22}$$

The general HF scheme, described in Sec. **??**, is iterative and aims to use these equations iteratively to find a stable solution for these parameters.

Off-diagonal terms. Consider the off-diagonal terms of Eq. (1.16). These contribute instead to the full gap, being out of diagonal in the 2×2 hamiltonian matrix. In terms of the pseudospin picture, they account for a transverse field. Define $\mathbf{q}$, $\mathbf{q}'$ as in Fig. 1.2b, and rewrite as for Eq. (1.17)

$$-\frac{2V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_\sigma [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}$$
$$= -\sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \right] \times V \sum_{\mathbf{q}'} \varphi^{(\gamma)*}_{\mathbf{q}'} \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'+\boldsymbol{\pi}\sigma} \tag{1.23}$$

Proceed as done previously: define the expectation value

$$v_{\mathbf{q}\sigma} \equiv i \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \tag{1.24}$$

(which is just the AF order parameter with a complex factor) and its γ-wave component, with $\gamma \in \{s^*, d, p_x, p_y\}$:

$$\begin{aligned}
v_\sigma^{(\gamma)} &= \frac{i}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \\
&= \frac{i}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \left[\varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle + \varphi^{(\gamma)}_{\mathbf{q}+\boldsymbol{\pi}} \langle \hat{\psi}_{\mathbf{q}+\boldsymbol{\pi}\sigma} \rangle \right] \\
&= \frac{i}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{q}} \left[\langle \hat{\psi}_{\mathbf{q}\sigma} \rangle - \langle \hat{\psi}^\dagger_{\mathbf{q}\sigma} \rangle \right] \\
&= -\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{\Psi}^\dagger_{\mathbf{q}\sigma} \tau^y \hat{\Psi}_{\mathbf{q}\sigma} \rangle \\
&= \frac{1}{2L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{q}} \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{q}\sigma}\}}{\tilde{E}_{\mathbf{q}\sigma}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta, \tilde{\mu}]
\end{aligned} \tag{1.25}$$

where we made use of the relations of Tab. 1.1 as well as Eq. (1.4). We also used the obvious facts that $\langle\hat{\psi}_{\mathbf{q}+\boldsymbol{\pi}\sigma}\rangle = \langle\hat{\psi}^\dagger_{\mathbf{k}\sigma}\rangle$ and $\varphi^{(\gamma)}_{\mathbf{q}+\boldsymbol{\pi}} = -\varphi^{(\gamma)}_{\mathbf{q}}$. From the definitions of the structure factors of Tab. **??**, it's immediate to see that:

$$v^{(s^*)}_\sigma, v^{(d)}_\sigma \in \mathbb{R} \qquad v^{(p_x)}_\sigma, v^{(p_y)}_\sigma \in i\mathbb{R}$$

Getting back to Eq. (1.23) and inserting these components, we get

$$-\frac{2V}{L_xL_y}\sum_{\mathbf{q},\mathbf{q}'}\sum_\sigma[\cos(2k_x)+\cos(2k_y)]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma} = iV\sum_{\gamma,\sigma}v^{(\gamma)}_\sigma\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{q}}\hat{\psi}_{\mathbf{q}\sigma} \tag{1.26}$$

We can expand on the MBZ as

$$-\frac{2V}{L_xL_y}\sum_{\mathbf{q},\mathbf{q}'}\sum_\sigma[\cos(2k_x)+\cos(2k_y)]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma} = iV\sum_{\gamma,\sigma}\sum_{\mathbf{q}\in\mathrm{MBZ}}\left[v^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}}\hat{\psi}_{\mathbf{q}\sigma}+v^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}+\boldsymbol{\pi}}\hat{\psi}^\dagger_{\mathbf{q}\sigma}\right] \tag{1.27}$$

Recalling Tab. **??**, for the symmetries we are considering

$$iv^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}} \in i\mathbb{R}$$

Moreover, since $\varphi^{(\gamma)}_{\mathbf{q}} = -\varphi^{(\gamma)}_{\mathbf{q}+\boldsymbol{\pi}}$, we have

$$iv^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}+\boldsymbol{\pi}} = \left(iv^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}}\right)^*$$

Therefore

$$-\frac{2V}{L_xL_y}\sum_{\mathbf{q},\mathbf{q}'}\sum_\sigma[\cos(2k_x)+\cos(2k_y)]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma} = V\sum_{\gamma,\sigma}\sum_{\mathbf{q}\in\mathrm{MBZ}}\left[iv^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}}\hat{\psi}_{\mathbf{q}\sigma}+\mathrm{h.c.}\right] \tag{1.28}$$

Four more real HFPs are then identified, with their self-consistency equations,

$$v^{(s^*)}_\sigma = \frac{1}{2L_xL_y}\sum_{\mathbf{q}\in\mathrm{MBZ}}(\cos q_x+\cos q_y)\frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{q}\sigma}\}}{\tilde{E}_{\mathbf{q}\sigma}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta,\tilde{\mu}] \tag{1.29}$$

$$v^{(d)}_\sigma = \frac{1}{2L_xL_y}\sum_{\mathbf{q}\in\mathrm{MBZ}}(\cos q_x-\cos q_y)\frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{q}\sigma}\}}{\tilde{E}_{\mathbf{q}\sigma}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta,\tilde{\mu}] \tag{1.30}$$

$$-iv^{(p_x)}_\sigma = \frac{1}{\sqrt{2}L_xL_y}\sum_{\mathbf{q}\in\mathrm{MBZ}}\sin q_x\frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{q}\sigma}\}}{\tilde{E}_{\mathbf{q}\sigma}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta,\tilde{\mu}] \tag{1.31}$$

$$-iv^{(p_y)}_\sigma = \frac{1}{\sqrt{2}L_xL_y}\sum_{\mathbf{q}\in\mathrm{MBZ}}\sin q_y\frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{q}\sigma}\}}{\tilde{E}_{\mathbf{q}\sigma}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta,\tilde{\mu}] \tag{1.32}$$

As we will discuss later in Sec. 1.4.2, not all these HFPs can coexist simultaneously. For now, let us move to the free energy density computation.

1.3.2 Fock contraction energy

Let us now get back to $E^{(\mathrm{AF})}_{\mathrm{F}/V}$, the double counting energy of Eq. (1.15). As said before, its definition is the same as for the normal phase, Eq. (**??**), with the averaging operation being performed on the AF ground state:

$$E^{(\mathrm{AF})}_{\mathrm{F}/V} \equiv \frac{V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\sum_\sigma[\cos(\delta k_x)+\cos(\delta k_y)]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\rangle\langle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}\rangle \tag{1.33}$$

Proceeding exactly as in Eq. (1.16) and making the change of variables of Fig. 1.2, we get

$$E_{\mathrm{F}/V}^{(\mathrm{AF})} = \frac{V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_{\sigma} \left[\cos\left(\delta q_x\right) + \cos\left(\delta q_y\right)\right] \Big[\underbrace{\langle \hat{n}_{\mathbf{q}\sigma}\rangle\langle \hat{n}_{\mathbf{q}'\sigma}\rangle}_{\text{Diagonal terms}} - \underbrace{\langle \hat{\psi}_{\mathbf{q}\sigma}\rangle\langle \hat{\psi}_{\mathbf{q}'\sigma}\rangle}_{\text{Off-diagonal terms}} \Big]$$

Apply the decomposition of Eq. (**??**) and use Eqns. (1.18), (1.24),

$$E_{\mathrm{F}/V}^{(\mathrm{AF})} = \frac{V}{2L_x L_y} \sum_{\gamma,\sigma} \sum_{\mathbf{q},\mathbf{q}'} \varphi_{\mathbf{q}}^{(\gamma)} \varphi_{\mathbf{q}'}^{(\gamma)*} \left[u_{\mathbf{q}\sigma} u_{\mathbf{q}'\sigma} + v_{\mathbf{q}\sigma} v_{\mathbf{q}'\sigma}\right] \tag{1.34}$$

Now: it holds

$$u_{\mathbf{q}\sigma} = \langle \hat{n}_{\mathbf{q}\sigma}\rangle \quad \Longrightarrow \quad u_{\mathbf{q}\sigma} = u_{\mathbf{q}\sigma}^*$$

and then

$$\begin{aligned}
\frac{1}{(L_x L_y)^2} \sum_{\mathbf{q},\mathbf{q}'} \varphi_{\mathbf{q}}^{(\gamma)} \varphi_{\mathbf{q}'}^{(\gamma)*} u_{\mathbf{q}\sigma} u_{\mathbf{q}'\sigma} &= \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} u_{\mathbf{q}\sigma}\right] \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}'\in\mathrm{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} u_{\mathbf{q}'\sigma}^*\right] \\
&= \left|u_{\sigma}^{(\gamma)}\right|^2
\end{aligned}$$

At the same time,

$$v_{\mathbf{q}\sigma} = i\langle \hat{\psi}_{\mathbf{q}\sigma}\rangle \quad \Longrightarrow \quad v_{\mathbf{q}+\boldsymbol{\pi}\sigma} = -v_{\mathbf{q}\sigma}^*$$

Since the form factors satisfy $\varphi_{\mathbf{q}+\boldsymbol{\pi}}^{(\gamma)} = -\varphi_{\mathbf{q}}^{(\gamma)}$, we get:

$$\begin{aligned}
\frac{1}{(L_x L_y)^2} \sum_{\mathbf{q},\mathbf{q}'} \varphi_{\mathbf{q}}^{(\gamma)} \varphi_{\mathbf{q}'}^{(\gamma)*} v_{\mathbf{q}\sigma} v_{\mathbf{q}'\sigma} &= \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} v_{\mathbf{q}\sigma}\right] \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}'\in\mathrm{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} v_{\mathbf{q}'\sigma}\right] \\
&= v_{\sigma}^{(\gamma)} \left[\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi_{\mathbf{k}+\boldsymbol{\pi}}^{(\gamma)*} v_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right] \\
&= v_{\sigma}^{(\gamma)} \left[\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi_{\mathbf{k}}^{(\gamma)*} v_{\mathbf{k}\sigma}^*\right] \\
&= v_{\sigma}^{(\gamma)} \times v_{\sigma}^{(\gamma)*}
\end{aligned}$$

where in the second passage we employed the change of variables $\mathbf{q}' = \mathbf{k}+\boldsymbol{\pi}$ and used BZ periodicity to map the original sum onto a new sum for $\mathbf{k} \in \mathrm{BZ}$. Inserting the above results in Eq. (1.34), this last becomes:

$$E_{\mathrm{F}/V}^{(\mathrm{AF})} = L_x L_y \times \frac{V}{2} \sum_{\gamma,\sigma} \left[\left|u_{\sigma}^{(\gamma)}\right|^2 + \left|v_{\sigma}^{(\gamma)}\right|^2\right] \tag{1.35}$$

Interestingly enough, the new HFPs amplitudes contribute to lowering the contraction energy.

1.4 Full MFT hamiltonian and gap complexification

Finally, it is time to collect everything in a single hamiltonian operator. The MFT model is approximated by:

$$\begin{aligned}
\hat{H} \simeq \sum_{\sigma} \sum_{\mathbf{k}\in\mathrm{MBZ}} \tilde{\epsilon}_{\mathbf{k}\sigma} \left[\hat{n}_{\mathbf{k}\sigma} - \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right] \\
- \sum_{\sigma} \sum_{\mathbf{k}\in\mathrm{MBZ}} \left[\tilde{\Delta}_{\mathbf{k}\sigma} \hat{\psi}_{\mathbf{k}\sigma} + \tilde{\Delta}_{\mathbf{k}\sigma}^* \hat{\psi}_{\mathbf{k}\sigma}^\dagger\right] - \left[E_{\mathrm{H}/U}^{(\mathrm{AF})} + E_{\mathrm{H}/V}^{(\mathrm{AF})} + E_{\mathrm{F}/V}^{(\mathrm{AF})}\right]
\end{aligned}$$

where the renormalized bands are given by:

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = \epsilon_{\mathbf{k}\sigma} + V \sum_{\gamma\in\{s^*,d\}} u_{\sigma}^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*}$$

and the full gap function by

$$\tilde{\Delta}_{\mathbf{k}\sigma} = \Delta_{\mathbf{k}\sigma} + iV \sum_{\gamma,\sigma} v_\sigma^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*}$$

The gap correction is purely imaginary, compared to the HM gap $\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU$. This has two obvious consequences: being no other contribution to the x pseudospin projections other than the pure HM ones, spin antisymmetry is preserved within such direction

$$\langle \hat{s}^x_{\mathbf{k}\sigma} \rangle = - \langle \hat{s}^x_{\mathbf{k}\bar{\sigma}} \rangle$$

which is relevant in the computation of the physical magnetization,

$$\begin{aligned} m &= \frac{1}{2L_x L_y} \sum_{\mathbf{r}} (-1)^{x+y} \langle \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow} \rangle \\ &= \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \left\langle \hat{s}^x_{\mathbf{k}\uparrow} - \hat{s}^x_{\mathbf{k}\downarrow} \right\rangle \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \left\langle \hat{s}^x_{\mathbf{k}\uparrow} \right\rangle \end{aligned}$$

A proof of the above derivation is given in Sec. A.1.6. Moreover, the net effect $\hat{H}_V$ is a "gap complexification". In other words,

$$\left| \mathrm{Im}\{\Delta_{\mathbf{k}\sigma}\} \right| \geq 0 \quad \Longrightarrow \quad \left| \langle s^y_{\mathbf{k}\sigma} \rangle \right| \geq 0$$

while for the HM the y pseudospin component was identically null.

1.4.1 The relative phase in the zero temperature half-filled ground state

In Sec. A.1.3, we derive analytically the AF ground state on the conventional HM at zero temperature, as a simple Bloch's transform of a polarized (tilted) pseudospins collective pure state. Eq. (A.27) reports the final form of said ground-state. Said quantum state is in general pretty similar to the conventional BCS state.

At half-filling and zero temperature, it is rather easy to extended that state to the new complexified gap. By performing an identical calculation as in Sec. A.1.3 we here omit, it's easy to show that the new ground state for the EHM in the AF phase is identical, with RMPs in the various amplitudes and a relative phase taking care of the gap function complex phase,

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k}\in\mathrm{MBZ}} \left[\sin\tilde{\theta}_{\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{k}\sigma} + e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}} \cos\tilde{\theta}_{\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma} \right] |\Omega_{\mathbf{k}\sigma}\rangle \tag{1.36}$$

which is identical to Eq. (A.27). The angles appearing in the above Eq. (1.8), which acquire a physical meaning: [To be continued...]

1.4.2 AF phase flavours and spin degeneracy

The AF phase we are studying cannot break inversion symmetry, a core property of the system which is respected from all our approximations. This translates in the simple requirement that:

$$\tilde{E}_{\mathbf{k}\sigma} \overset{!}{=} \tilde{E}_{-\mathbf{k}\sigma}$$

We know that:

$$\tilde{E}_{\mathbf{k}\sigma} = \sqrt{\tilde{\epsilon}^2_{\mathbf{k}\sigma} + \mathrm{Re}^2\{\tilde{\Delta}_{\mathbf{k}\sigma}\} + \mathrm{Im}^2\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}$$

The renormalized bare bands as well as the real part of the gap satisfy the inversion symmetry requirement,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = \tilde{\epsilon}_{-\mathbf{k}\sigma} \qquad \mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} = \mathrm{Re}\{\tilde{\Delta}_{-\mathbf{k}\sigma}\}$$

hence inversion symmetry is respected only if

$$\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} = \pm \mathrm{Im}\{\tilde{\Delta}_{-\mathbf{k}\sigma}\}$$

Phase	HFPs
sAF	$\{m, u^{(s^*)}, u^{(d)}, v^{(s^*)}, v^{(d)}\}$
aAF	$\{m, u^{(s^*)}, u^{(d)}, -iv^{(p_x)}, -iv^{(p_y)}\}$

Table 1.2 Set of HFP for the two possible realization of the AF phase, respectively a symmetric and an antisymmetric imaginary part of the gap.

which is: either the imaginary part of the gap is inversion symmetric, or is antisymmetric. Based on this trivial consideration, we are able two discard two of the four self-consistency equations for the v-related HFPs: s^* and d-wave symmetries HFPs – Eqns. (1.29) and (1.30), respectively – are nonzero if $\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}$ is momentum-even; p_x and p_y-wave symmetries HFPs – Eqns. (1.31) and (1.32), respectively – are nonzero if $\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}$ is momentum-odd. The two option cannot coexist, and separate simulations can be implemented in the two symmetry sectors. We name the two possible symmetries "sAF" (symmetric antiferromagnet) and "aAF" (antisymmetric antiferromagnet). The set of HFPs then restricts in these AF structures, and is reported for both cases in Tab. 1.2.

Moreover, from now on we will omit the σ index, working just in the $\uparrow$ sector. As said in Sec. **??**, the AF phase does not break $\mathbb{Z}_2$ spin inversion symmetry, being the SDW ordering generated by a staggered but symmetric spin density distribution among sites. The MFT hamiltonian is invariant by a $\sigma \leftrightarrow \overline{\sigma}$ exchange, thus all measurable physical quantities must be. What is unmeasurable and has no effect on Physics is free to be σ-dependent, as for the sign of $\mathrm{Re}\{\Delta_{\mathbf{k}\sigma}\}$. Since all it's relevant of both the complex components of the gap function is their magnitude, we can safely treat σ as a pure degeneracy. This gives a simple equation for the last HFP,

$$\begin{aligned} m &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \langle \hat{s}^x_{\mathbf{k}\uparrow} \rangle \\ &= -\frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta, \tilde{\mu}] \end{aligned} \tag{1.37}$$

[To be continued...]

1.5 Stability of the AF phase

The discussion above is formally correct and self-consistent, thus a HF algorithm can be sketched to extract the HFPs. However we cannot proceed without considering the instability effects of Sec. A.1.7. As is known (and we would have liked to notice earlier) the AF-SSB at wavevector $\boldsymbol{\pi}$ we are studying here, responsible of halving the BZ down to the MBZ, can produce a stable phase on pure Hubbard lattices only at half filling [**singh1990collective**]. For the EHM the situation is even worse than that: repeating the simple Linear Reponse Theory (LRT) computations of Sec. A.1.7 and mimicking Eq. (A.32), we get a proper response function

$$\tilde{\chi}_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k}, \omega) = \frac{\chi_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k}, \omega)}{1 - [U - 2V(\cos k_x + \cos k_y)]\, \chi_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k}, \omega)} \tag{1.38}$$

[Double check the red part and conclude this paragraph...].

As is well documented in literature [**wietek2021stripes**], AF can establish in a wide range of spatial structures with particular shapes, for example stripes. In order to extract the precise AF phase, we should

1. Choose an initializer wavevector, say $\mathbf{k} = \boldsymbol{\pi}$;
2. Solve self-consistently for the HFPs at $\mathbf{k}$;
3. Compute the free energy F as a functional of the wavevector around $\mathbf{k}$;
4. Look for gradient descent of the free energy, $\boldsymbol{\nabla} F < 0$;

5. If the gradient descent is found, update $\mathbf{k} \to \mathbf{k} + \delta\mathbf{k}$ and repeat from step 2, otherwise halt.

Of course the sketched procedure can lead to a lot of complications, first the fact that at incommensurate wavevector $\mathbf{k} \neq \boldsymbol{\pi}$ the unit cell is not as simple as a pair of sites, but can become much larger requiring more refined computational solutions.

In this text we will show the self consistent result for AF-SSB at wavevector $\boldsymbol{\pi}$ even at finite doping as a first step in the above sketched procedure. The presented results are self-consistent, but unfortunately deeply unstable and unphysical at finite doping, a situation where an incommensurate and insulating AF phase is preferred to this commensurate metallic counterpart. Nevertheless we present them anyway.

1.6 AF free energy density

In order to derive the expression of the free energy density for the AF phase, we proceed precisely as in Sec. **??**. The only notable difference is in entropy: Eq. (**??**) is written for two opposite bands on MBZ

$$s = s^{(+)} + s^{(-)}$$

with

$$s^{(\pm)} \equiv -\frac{k_B}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \Big[\Big(1 - f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) \ln \Big(1 - f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) + f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) \ln f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) \Big] \quad (1.39)$$

Consider also that we are computing free energy on the gran-canonical hamiltonian

$$\hat{H} - \mu\hat{N} \simeq \sum_{\sigma} \sum_{\mathbf{k}\in\mathrm{MBZ}} \Big[(\tilde{E}_{\mathbf{k}} - \tilde{\mu}) \hat{\gamma}^{(+)\dagger}_{\mathbf{k}\sigma} \hat{\gamma}^{(+)}_{\mathbf{k}\sigma} - (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) \hat{\gamma}^{(-)\dagger}_{\mathbf{k}\sigma} \hat{\gamma}^{(-)}_{\mathbf{k}\sigma} \Big] + E_{\mathrm{c}}^{(\mathrm{AF})}$$

being $E_{\mathrm{c}}^{(\mathrm{AF})}$ the total AF contraction energy. Note that on the left-hand side we have μ and on the right-hand side its RMP counterpart $\tilde{\mu}$.

With identical meaning of the notation with respect to Sec. **??**, the free energy contributions are:

1. Quasiparticles energy, taking care of spin degeneracy:

$$e_{\mathrm{g}}^{(\mathrm{AF})} = \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) - \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})$$

2. Contraction energy:

$$e_{\mathrm{c}}^{(\mathrm{AF})} = U(m^2 - n^2) + V \left[2zn^2 - \sum_{\gamma} \Big(\big|u^{(\gamma)}\big|^2 + \big|v^{(\gamma)}\big|^2 \Big) \right]$$

3. Entropic energy

$$e_{s}^{(\mathrm{AF})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \ln \Big(1 - f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) - \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})$$
$$\frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \ln \Big(1 - f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) + \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})$$

4. Chemical potential correction

$$e_{n}^{(\mathrm{AF})} = \mu n$$

Compose everything

$$f_{\mathrm{MFT}}^{(\mathrm{AF})} \equiv e_{\mathrm{g}}^{(\mathrm{AF})} + e_{\mathrm{c}}^{(\mathrm{AF})} + e_{s}^{(\mathrm{AF})} + e_{n}^{(\mathrm{AF})}$$

and we get

$$f_{\mathrm{MFT}}^{(\mathrm{AF})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \left[\ln\left(1 - f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\right) + \ln\left(1 - f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\right)\right] + U(m^2 - n^2) + V\left[2zn^2 - \sum_{\gamma}\left(\left|u^{(\gamma)}\right|^2 + \left|v^{(\gamma)}\right|^2\right)\right] + \mu n \quad (1.40)$$

It's of particular interest the half-filling condition, with $\tilde{\mu} = 0$. [To be continued...]

1.7 Numerical simulations

[To be continued...]

1.7.1 HF iterative algorithm sketch

[To be continued...]

Chapter 2

Superconducting instability

This chapter is devoted to studying the superconducting phase of the system. The only symmetry we assume to break is the $\mathrm{U}^{\mathrm{c}}(1)$ charge symmetry, thus allowing for superconducting fluctuations. As is described thoroughly in Sec. 2.3.1, the hopping amplitude is renormalized because of the non-local attraction. The symmetry structure of the pairing mechanism determines the contributing Cooper fluctuations: for s-wave and d-wave superconductivity, only the o.s. Cooper term contributes; for p_ℓ-wave superconductivity, the s.s. term contributes as well. In the following sections, a derivation containing both Cooper terms is proposed.

2.1 The SC phase and its symmetries

The starting point is, as done for the other phases, the discussion of the symmetries of the SC phase. We said in introduction we are studying the

$$\mathrm{SU}(2) \times \mathrm{U}^{\mathrm{c}}(1) \times (\text{Point group}) \quad \overset{\text{SSB}}{\to} \quad \mathrm{SU}(2) \times (\text{Point group})$$

symmetry breaking. Therefore the system density we assume to be uniform across all sites, similarly to the normal phase discussed in Chap. **??**. [To be continued...]

Let us highlight now an important point regarding the Cooper class Wick's contractions performed on the general EHM of Eq. (**??**),

$$\hat{H} = \underbrace{-t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma}}_{\hat{H}_t} \underbrace{+U \sum_{i} \hat{c}^\dagger_{i\uparrow} \hat{c}^\dagger_{i\downarrow} \hat{c}_{i\downarrow} \hat{c}_{i\uparrow}}_{\hat{H}_U} \underbrace{-V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{c}^\dagger_{i\sigma} \hat{c}^\dagger_{j\sigma'} \hat{c}_{j\sigma'} \hat{c}_{i\sigma}}_{\hat{H}_V}$$

As is discussed in Sec. **??**, when applying Wick's theorem the resulting terms break the natural symmetries of the model. The superconducting symmetry we study breaks just the $\mathrm{U}^{\mathrm{c}}(1)$ charge conservation. Now, when dealing with superconducting Cooper pairing we need to account also for the spatial structure of the Cooper pair itself. Consider the generic Cooper fluctuation

$$\langle \hat{c}^\dagger_{i\sigma} \hat{c}^\dagger_{j\sigma'} \rangle \quad \text{with } i, j \text{ NN}$$

From basic Quantum Mechanics we know the summation rules of the spin algebra $\mathfrak{su}(2)$,

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$$

The two pairing channels are, at this level, the singlet channel associated to total spin 0 and the triplet channel associated to total spin 1. If we impose a specific spatial symmetry on the hamiltonian the ground state wavefunction will follow naturally, and the pairing channel will be the one providing a total anti-symmetry to the full wavefunction. This gives a selection rule over the relevant pairings: if we work with space symmetric structures – say, s^*-wave or d-wave – the pairing will take place in singlet channel, allowing us to eliminate the triplet pairing. Working with antisymmetric structures, inversely, we can eliminate the singlet channel. This concept is summarized in Tab. 2.1.

Spatial structure	Pairing channel	RWCs in Cooper channel
Symmetric wave function	Singlet pairing	Just $\langle \hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\overline{\sigma}}\rangle$
Anti-symmetric wave function	Triplet pairing	Both $\langle \hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\overline{\sigma}}\rangle$ and $\langle \hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\sigma}\rangle$

Table 2.1 Relation of the Cooper pairing channel with the wavefunction spatial symmetry (intended as the inversion $(x, y) \to (-x, -y)$).

2.1.1 Superconductivity in the HM

In Sec. A.2, we discuss in detail the (impossibility of a stable) s-wave SC phase in the conventional HM. This is an obvious result, since the two-body interaction thereby present $\hat{H}_U$ is local and repulsive. We nevertheless treat the model within the MFT framework, and here recall the main widely known results, and later use them as groundwork for the EHM extension. Similarly to Sec. 1.1.1, let us define the operators

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} \tag{2.1}$$

$$\hat{\phi}_{\mathbf{k}} \equiv \hat{c}^\dagger_{\mathbf{k}\uparrow}\hat{c}^\dagger_{-\mathbf{k}\downarrow} \tag{2.2}$$

whose EVs are relevant for the SC phase. Let:

$$u_{\mathbf{q}\sigma} \equiv \langle \hat{n}_{\mathbf{q}\sigma}\rangle \tag{2.3}$$

$$w_{\mathbf{q}\sigma} \equiv \langle \hat{\phi}^\dagger_{\mathbf{q}}\rangle \tag{2.4}$$

The MFT hamiltonian is given explicitly for the HM in Eq. (A.43),

$$\hat{H}_t + \hat{H}_U \overset{\text{MFT}}{\simeq} \sum_{\mathbf{k}\in\text{BZ}}\sum_{\sigma}\epsilon_{\mathbf{k}\sigma}\left[\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} - \hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right] - \sum_{\mathbf{q}\in\text{BZ}}\left[\Delta^{(s)}\hat{\phi}_{\mathbf{q}} + \Delta^{(s)*}\hat{\phi}^\dagger_{\mathbf{q}}\right] + nU\times\hat{N} - \left[E^{(\text{SC})}_{\text{H}/U} + E^{(\text{SC})}_{\text{C}/U}\right]$$

with $E^{(\text{SC})}_{\text{H}/U}$, $E^{(\text{SC})}_{\text{C}/U}$ the contraction energies arising from Hartree and Cooper channels, respectively in Eq. (A.36) and (A.40). In deriving this result we decomposed the interaction of the intermediate passage of Eq. (A.37) and assumed the Cooper pair to have null total momentum. Moreover, we recognized in the passage of Eq. (A.39) the presence of a s-wave harmonic transform of the function $w_{\mathbf{q}}$, and thus defined in Eq. (A.41),

$$w^{(s)} \equiv \frac{1}{L_x L_y}\sum_{\mathbf{q}\in\text{BZ}} w_{\mathbf{q}} \quad \text{and} \quad \Delta^{(s)} \equiv -U w^{(s)}$$

Nambu-Bogoliubov representation of the MFT hamiltonian is given in Eq. (A.46) and equivalently in Eq. (**??**),

$$\hat{H}_t + \hat{H}_U - \mu\hat{N} \simeq \sum_{\mathbf{k}\in\text{BZ}}\sum_{\sigma}\hat{\Phi}^\dagger_{\mathbf{k}} h_{\mathbf{k}}\hat{\Phi}_{\mathbf{k}} - \left[E^{(\text{SC})}_{\text{A}/t} + E^{(\text{SC})}_{\text{H}/U} + E^{(\text{SC})}_{\text{C}/U}\right]$$

being

$$h_{\mathbf{k}} \equiv \begin{bmatrix} \xi_{\mathbf{k}} & -\Delta_{\mathbf{k}} \\ -\Delta^*_{\mathbf{k}} & -\xi_{\mathbf{k}} \end{bmatrix} \quad \text{and} \quad \xi_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} - \tilde{\mu}$$

and $\Delta_{\mathbf{k}} = \Delta^{(s)}$. Nambu spinors are, in this case,

$$\hat{\Phi}_{\mathbf{k}} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\uparrow} \\ \hat{c}^\dagger_{-\mathbf{k}\downarrow} \end{bmatrix}$$

For the SC phase in the HM model, in general, we can admit for a complex gap function – although being the gap a constant it can always be reduced to a purely real gap. Diagonalization by the angles defined in Eq. (A.47) is performed via the composed rotation of Eq. (A.49),

$$W_{\mathbf{k}} = \begin{bmatrix} \cos\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & -\sin\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \\ \sin\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & \cos\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \end{bmatrix}$$

EV	Meaning	Equation
$\langle \hat{s}^x_{\mathbf{k}} \rangle$	$\left\langle \hat{\phi}_{\mathbf{k}} + \hat{\phi}^\dagger_{\mathbf{k}} \right\rangle$	(A.55)
$i \langle \hat{s}^y_{\mathbf{k}} \rangle$	$\left\langle \hat{\phi}_{\mathbf{k}\sigma} - \hat{\phi}^\dagger_{\mathbf{k}} \right\rangle$	(A.56)
$\langle \hat{s}^z_{\mathbf{k}} \rangle$	$\langle \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{\mathbf{k}\downarrow} \rangle - 1$	(A.57)

Table 2.2 List of the various expectation values (EVs) of each pseudospin component, their meaning in terms of fermionic operators and the reference equations from the pure HM. The z component refers also to Eq. (A.44).

The Bogoliubov bands are $\pm E_{\mathbf{k}} = \sqrt{\xi^2_{\mathbf{k}} + |\Delta_{\mathbf{k}}|^2}$. The only crucial difference with the AF Bogoliubov bands it that here, apart from the absence of the spin index – a degeneracy that is solved now by enlarging the computation to the entire BZ –, also the bare bands $\epsilon_{\mathbf{k}}$ are substituted by their shifted version $\xi_{\mathbf{k}}$. This difference is essential: it is this feature that makes the superconducting phase gapped in an insulating sense, which means, opening the gap at the position of $\tilde{\mu}$. Instead for the AF phase the gap was always opened at "midband", the MBZ contour where $\epsilon_{\mathbf{k}} = 0$. As a final remark, as is proven trivially at the end of Sec. A.2.5, s-wave SC is impossible at MFT level on the pure HM. This was obvious, due to the necessity of an even small attractive interaction to form Cooper pairs. To conclude, in Tab. 2.2 we report the operator-pseudospin map for each direction.

2.1.2 Extension to the EHM

Add now the non-local interaction $\hat{H}_V$: translational invariance is preserved while charge conservation is broken, thus all three channels (Hartree, Fock and Cooper) contribute via RWCs to the MFT hamiltonian. As for the AF phase, we aim to derive a MFT solution for the SC phase which is an extension of the HM solution of Sec. A.2, with an effective redefinition of the various physical quantities and mathematical objects,

$$\xi_{\mathbf{k}} \to \tilde{\xi}_{\mathbf{k}} \qquad E_{\mathbf{k}} \to \tilde{E}_{\mathbf{k}} \qquad \Delta_{\mathbf{k}} \to \tilde{\Delta}_{\mathbf{k}} \qquad \mu \to \tilde{\mu}$$

To be completely correct, the chemical potential was already renormalized in the HM derivation in Sec. A.2.1. It here acquires yet another renormalization. The band energies renormalization is simply

$$\tilde{E}_{\mathbf{k}} \equiv \sqrt{\tilde{\xi}^2_{\mathbf{k}} + |\tilde{\Delta}_{\mathbf{k}}|^2}$$

and as for the AF phase:

$$\langle \hat{\Phi}^\dagger_{\mathbf{k}} \tau^x \hat{\Phi}_{\mathbf{k}} \rangle = -\sin(2\tilde{\theta}_{\mathbf{k}}) \sin(2\tilde{\zeta}_{\mathbf{k}}) \langle \hat{\Gamma}^\dagger_{\mathbf{k}} \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \tag{2.5}$$

$$\langle \hat{\Phi}^\dagger_{\mathbf{k}} \tau^y \hat{\Phi}_{\mathbf{k}} \rangle = -\sin(2\tilde{\theta}_{\mathbf{k}}) \cos(2\tilde{\zeta}_{\mathbf{k}}) \langle \hat{\Gamma}^\dagger_{\mathbf{k}} \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \tag{2.6}$$

$$\langle \hat{\Phi}^\dagger_{\mathbf{k}} \tau^z \hat{\Phi}_{\mathbf{k}} \rangle = \cos(2\tilde{\theta}_{\mathbf{k}}) \langle \hat{\Gamma}^\dagger_{\mathbf{k}} \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \tag{2.7}$$

These equations are identical to the ones for the pure HM, Eqns. (A.55), (A.56) and (A.57). In the tilted frame, we have the analogous of Eq. (1.6) with the important exception that the chemical potential in the Fermi-Dirac factors is zero,

$$\langle \hat{\Gamma}^\dagger_{\mathbf{k}} \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle = \frac{1}{2} \left[f\left(\tilde{E}_{\mathbf{k}}; \beta, 0\right) - f\left(-\tilde{E}_{\mathbf{k}}; \beta, 0\right) \right] \tag{2.8}$$

This is because all the chemical potential operatorial contributions were absorbed in the bare bands $\tilde{\xi}_{\mathbf{k}}$, thus the pseudospin state is thermally described by an effective null chemical potential.

The (renormalized) thermal factors as in Eq. (A.54) are

$$\tilde{\mathrm{Th}}^{(\pm)}_{\mathbf{k}}[\beta, \tilde{\mu}] \equiv f\left(\tilde{E}_{\mathbf{k}}; \beta, 0\right) \pm f\left(-\tilde{E}_{\mathbf{k}}; \beta, 0\right) \tag{2.9}$$

which simply reduce to

$$\tilde{\mathrm{Th}}_{\mathbf{k}}^{(+)} = 1$$
$$\tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)} = -\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)$$

For the angles it holds

$$\sin(2\tilde{\xi}_{\mathbf{k}}) = \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{|\tilde{\Delta}_{\mathbf{k}}|} \quad \cos(2\tilde{\xi}_{\mathbf{k}}) = \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{|\tilde{\Delta}_{\mathbf{k}}|} \quad \sin(2\tilde{\theta}_{\mathbf{k}}) = \frac{|\tilde{\Delta}_{\mathbf{k}}|}{\tilde{E}_{\mathbf{k}}} \quad \cos(2\tilde{\theta}_{\mathbf{k}}) = \frac{\tilde{\xi}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tag{2.10}$$

From Tabs. **??** and **??**, we know all channel contribute. It is already known the effect of the Fock contractions, responsible as for the normal phase (Chap. **??**) of the bare bands renormalization. Cooper terms are to be handled more carefully, since can contribute to superconductivity both in the s.s. and in the o.s. spin sectors.

2.2 Non-local Hartree effects: chemical potential renormalization

As always, let us start with a very brief section about the Hartree effects on the MFT decomposition of the hamiltonian. We will not delve in the detailed calculations of the Hartree channel MFT effects, already described in previous chapter as well as in Sec. A.2. The only effect here is the chemical potential renormalization.

2.2.1 Hartree channel in the SC phase: analytical derivation

The Hartree fluctuations are density-density couplings in the MFT hamiltonian. Knowing the deal with a uniform-density phase, we can re-use as it is the derivation of Sec. **??**. Without repeating the calculations, the $\hat{H}_V$ is reduced in the Hartree channel to

$$\hat{H}_V \simeq -2nzV \times \hat{N} - E_{\mathrm{H}/V}^{(\mathrm{SC})} + (\text{all the rest})$$

This, together with the $\hat{H}_U$ Hartree channel decomposition

$$\hat{H}_U \simeq nU \times \hat{N} - E_{\mathrm{H}/U}^{(\mathrm{SC})}$$

gives the chemical potential full renormalization, identical to Eq. (**??**),

$$\tilde{\mu} \equiv \mu + n(2zV - U) \tag{2.11}$$

2.2.2 Hartree contraction energy

The contraction energy for the non-local sector was derived in Eq. (**??**),

$$E_{\mathrm{H}/V}^{(\mathrm{SC})} = -L_x L_y \times 2zn^2V$$

and together with the local counterpart of Eq. (**??**)

$$E_{\mathrm{H}/U}^{(\mathrm{SC})} = L_x L_y \times n^2U$$

Now we move to the more interesing Fock sector.

2.3 Fock effects: bare bands renormalization

[To be continued...]

2.3.1 Fock channel in the SC phase: analytical derivation

When it comes to the Fock RWCs, being the SC phase density-uniform we can re-use almost the entirety of the derivation of Sec. **??**. Setting as in Eq. (**??**)

$$u_{\mathbf{q}\sigma} \equiv \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle \tag{2.12}$$

Eq. (**??**) stands perfectly valid, translated to the SC phase

$$\begin{aligned} V \sum_{\langle ij \rangle} \sum_{\sigma} \Big[\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle \Big] \\ \overset{\mathcal{F}}{=} \sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} u_{\mathbf{q}\sigma} \right] \times V \sum_{\mathbf{q}' \in \mathrm{BZ}} \varphi^{(\gamma)*}_{\mathbf{q}'} \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} - E^{(\mathrm{SC})}_{\mathrm{F}/V} \end{aligned} \tag{2.13}$$

being

$$E^{(\mathrm{SC})}_{\mathrm{F}/V} \equiv \frac{V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_{\sigma} \left[\cos(\delta q_x) + \cos(\delta q_y) \right] \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle \langle \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} \rangle \tag{2.14}$$

To get to this point, we employed the assumption

$$\langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}'\sigma'} \rangle = \delta_{\mathbf{k}=\mathbf{k}'} \delta_{\sigma=\sigma'} \langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \rangle$$

which essentially states translational invariance and spin degeneracy. This assumption is obviously well-verified by the BCS ground state of Eq. [Missing]. Recognising the symmetry decomposition

$$u^{(\gamma)}_{\sigma} = \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle$$

due to inversion symmetry the conclusion is identical to Eqns. (**??**) and (**??**) (considering spin degeneracy),

$$\tilde{\epsilon}_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} + u^{(s^*)} V (\cos q_x + \cos q_y) + u^{(d)} V (\cos q_x - \cos q_y)$$

In order to estimate the γ-wave components renormalizing the bare bands, we need to compute separately in spin sectors the EVs of $u_{\mathbf{k}\sigma}$

$$\begin{aligned} u_{\mathbf{k}\uparrow} &= \langle \hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}_{\mathbf{k}\uparrow} \rangle \qquad (2.15) \\ &= \left\langle \hat{\Phi}^\dagger_{\mathbf{k}} \frac{\mathbb{I} + \tau^z}{2} \hat{\Phi}_{\mathbf{k}} \right\rangle \\ &= \frac{1}{2} \left[\left\langle \hat{\Gamma}^\dagger_{\mathbf{k}} \hat{\Gamma}_{\mathbf{k}} \right\rangle + \langle \hat{s}^z_{\mathbf{k}} \rangle \right] \\ &= \frac{1}{2} \left[\tilde{\mathrm{Th}}^{(+)}_{\mathbf{k}}[\beta] + \cos 2\tilde{\theta}_{\mathbf{k}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta] \right] \\ &= \frac{1 + \cos 2\tilde{\theta}_{\mathbf{k}}}{2} f\left(\tilde{E}_{\mathbf{k}}; \beta, 0\right) + \frac{1 - \cos 2\tilde{\theta}_{\mathbf{k}}}{2} f\left(-\tilde{E}_{\mathbf{k}}; \beta, 0\right) \\ &= \cos^2 \tilde{\theta}_{\mathbf{k}} f\left(\tilde{E}_{\mathbf{k}}; \beta, 0\right) + \sin^2 \tilde{\theta}_{\mathbf{k}} f\left(-\tilde{E}_{\mathbf{k}}; \beta, 0\right) \\ &= \sin^2 \tilde{\theta}_{\mathbf{k}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) + f\left(\tilde{E}_{\mathbf{k}}; \beta, 0\right) \qquad (2.16) \end{aligned}$$

having we used the simple relation

$$\begin{aligned} \frac{1}{e^{-x}+1} - \frac{1}{e^{x}+1} &= \frac{e^x - 1}{e^x + 1} \\ &= \tanh\left(\frac{x}{2}\right) \end{aligned}$$

For the spin $\downarrow$ EV,

$$
\begin{aligned}
u_{-\mathbf{k}\downarrow} &= \langle \hat{c}^\dagger_{-\mathbf{k}\downarrow} \hat{c}_{-\mathbf{k}\downarrow} \rangle \\
&= 1 - \langle \hat{c}_{-\mathbf{k}\downarrow} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \rangle \\
&= 1 - \left\langle \hat{\Phi}^\dagger_{\mathbf{k}} \frac{\mathbb{I} - \tau^z}{2} \hat{\Phi}_{\mathbf{k}} \right\rangle \\
&= 1 - \frac{1}{2} \left[\left\langle \hat{\Gamma}^\dagger_{\mathbf{k}} \hat{\Gamma}_{\mathbf{k}} \right\rangle - \langle \hat{s}^z_{\mathbf{k}} \rangle \right] \\
&= 1 - \frac{1}{2} \left[\tilde{\mathrm{Th}}^{(+)}_{\mathbf{k}}[\beta] - \cos 2\tilde{\theta}_{\mathbf{k}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta] \right] \\
&= 1 - \frac{1 - \cos 2\tilde{\theta}_{\mathbf{k}}}{2} f\left(\tilde{E}_{\mathbf{k}}; \beta, 0\right) - \frac{1 + \cos 2\tilde{\theta}_{\mathbf{k}}}{2} f\left(-\tilde{E}_{\mathbf{k}}; \beta, 0\right) \\
&= 1 - \sin^2 \tilde{\theta}_{\mathbf{k}} f\left(\tilde{E}_{\mathbf{k}}; \beta, 0\right) - \cos^2 \tilde{\theta}_{\mathbf{k}} f\left(-\tilde{E}_{\mathbf{k}}; \beta, 0\right) \\
&= 1 - \sin^2 \tilde{\theta}_{\mathbf{k}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) - f\left(-\tilde{E}_{\mathbf{k}}; \beta, 0\right) \\
&= -\sin^2 \tilde{\theta}_{\mathbf{k}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) + f\left(\tilde{E}_{\mathbf{k}}; \beta, 0\right)
\end{aligned}
\tag{2.17}
$$

These results will be used to extract self-consistency equations for each HFP. Note that

$$\lim_{|\Delta_{\mathbf{k}}| \to 0} u_{\mathbf{k}\sigma} = f(\tilde{\xi}_{\mathbf{k}}; \beta, 0)$$

Correctly, the null-gap limit reduces to a perfectly degenerate Fermi gas.

2.3.2 Fock contraction energy

In terms of the energy shift due to Wick's contraction, Eq. (**??**) holds identically

$$E^{(\mathrm{SC})}_{\mathrm{F}/V} = L_x L_y \times \frac{V}{2} \sum_{\sigma} \left[\left| u^{(s^*)} \right|^2 + \left| u^{(d)} \right|^2 \right]$$

with the only obvious difference that the $u^{(\gamma)}$ are computed on the SC ground state by the means of Eqns. (2.16), (2.17).

2.4 Cooper channel in the o.s. and s.s. sectors

Let us get back to the Fourier transform of Eq. (**??**)

$$\hat{H}_V \overset{\mathcal{F}}{=} -\frac{V}{L_x L_y} \sum_{\sigma\sigma'} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos(\delta k_x) + \cos(\delta k_y) \right] \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma'} \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}$$

which decomposes in s.s. and o.s. channels as:

$$
\begin{aligned}
\hat{H}_V \overset{\mathcal{F}}{=} & \overbrace{-\frac{V}{L_x L_y} \sum_{\sigma} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos(\delta k_x) + \cos(\delta k_y) \right] \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}^{\text{s.s.}} \\
& \underbrace{-\frac{V}{L_x L_y} \sum_{\sigma} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos(\delta k_x) + \cos(\delta k_y) \right] \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}} \hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}_{\text{o.s.}}
\end{aligned}
$$

Following Tab. 2.1, in the singlet pairing channel only the o.s. sector produces RWCs. The latter is made up of two identical terms: this is because it holds

$$\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}} \hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} = \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}} \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}$$

due to fermionic anti-commutation rules, and since the composite transform

$$\mathbf{k} \to -\mathbf{k} \qquad \mathbf{k}' \to -\mathbf{k}'$$

leaves the form factor $\cos(\delta k_x) + \cos(\delta k_y)$ invariant, we reduce the spin sum to two times the $\sigma = \uparrow$ part,

$$\hat{H}_V^{(\text{o.s.})} = -\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos(\delta k_x) + \cos(\delta k_y)\right] \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \tag{2.18}$$

[To be continued...]

2.5 Cooper effects in the singlet channel: gap renormalization

[To be continued...]

2.5.1 Cooper-singlet channel in the SC phase: analytical derivation

This section deals with Cooper fluctuations induced by the o.s. part of the non-local hamiltonian –referring to the notation of Eq. (**??**)– somewhat the simplest form of Cooper pairing. This sector contributes both to singlet pairing and triplet pairing. Considering Cooper fluctuations in the singlet channel, we need to break $\mathrm{U}^{\mathrm{c}}(1)$ symmetry imposing space inversion symmetry, while for the triplet anti-symmetry is required. Set as in Eq. (2.4) the EV

$$w_{\mathbf{q}} = \langle \hat{\phi}^\dagger_{\mathbf{q}} \rangle$$

Let us now break down the MFT discussion for the local and non-local interactions.

Local repulsion. The detailed treatment of $\hat{H}_U$ is discussed in Sec. A.2.2, in the context of conventional HM. We here just sketch the most important results. By contracting in the Cooper channel the Fourier transform of $\hat{H}_U$ of Eq. (**??**),

$$\hat{H}_U \overset{\mathcal{F}}{=} \frac{U}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow}$$

in the hypothesis that preserving space translations invariance the net momentum of the Cooper pair is null, as in Eq. (A.39) the local part of the hamiltonian is reduced to

$$\hat{H}_U \simeq (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \left[w_{\mathbf{q}} \hat{\phi}_{\mathbf{q}'} + w^*_{\mathbf{q}'} \hat{\phi}^\dagger_{\mathbf{q}} \right] - E^{(\text{SC})}_{\text{C}/U} \tag{2.19}$$

being $E^{(\text{SC})}_{\text{C}/U}$ the Cooper contraction given in Eq. (A.40). The Hartree channel is discussed in Sec. A.2.1. Let now:

$$w^{(s)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} w_{\mathbf{q}}$$

which reduces Eq. (2.19) to

$$\hat{H}_U \simeq (\text{Hartree channel}) + U \sum_{\mathbf{q} \in \text{BZ}} \left[w^{(s)} \hat{\phi}_{\mathbf{q}} + w^{(s)*} \hat{\phi}^\dagger_{\mathbf{q}} \right] - E^{(\text{SC})}_{\text{C}/U} \tag{2.20}$$

As in Eq. (A.42), this will produce the s-wave part of the gap function.

Non-local attraction. Consider Eq. (2.18), and perform Wick contractions, giving

$$\begin{aligned} \hat{H}_V^{(\text{o.s.})} \simeq -\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos(\delta k_x) + \cos(\delta k_y)\right] \times \Big[&\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \\ &+ \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \rangle - \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \rangle \langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \rangle \Big] + (\text{all the rest}) \end{aligned}$$

Identical considerations as above hold, and just the $\mathbf{K} = \mathbf{0}$ term contributes. We have finally

$$\hat{H}_V^{(\text{o.s.})} \simeq -\frac{2V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \left[\cos(\delta q_x) + \cos(\delta q_y)\right] \left[w_{\mathbf{q}} \hat{\phi}_{\mathbf{q}'} + w_{\mathbf{q}}^* \hat{\phi}_{\mathbf{q}'}^\dagger\right] - E_{\text{C}/V/\text{o.s.}}^{(\text{SC})} + (\text{all the rest}) \tag{2.21}$$

where the contraction energy is

$$E_{\text{C}/V/\text{o.s.}}^{(\text{SC})} \equiv -\frac{2V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \left[\cos(\delta q_x) + \cos(\delta q_y)\right] w_{\mathbf{q}} w_{\mathbf{q}'}^* \tag{2.22}$$

Making use of the decomposition of Eq. (**??**), Eq. (2.21) becomes

$$\begin{aligned}
\hat{H}_V^{(\text{o.s.})} &\simeq -V \sum_\gamma \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}} \varphi_{\mathbf{q}}^{(\gamma)} w_{\mathbf{q}}\right] \times \sum_{\mathbf{q}'} \varphi_{\mathbf{q}'}^{(\gamma)*} \hat{\phi}_{\mathbf{q}'} + \text{h.c.} - E_{\text{C}/V/\text{o.s.}}^{(\text{SC})} + (\text{all the rest}) \\
&= -V \sum_\gamma \sum_{\mathbf{q}\in\text{BZ}} w^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} \hat{\phi}_{\mathbf{q}} + \text{h.c.} - E_{\text{C}/V/\text{o.s.}}^{(\text{SC})} + (\text{all the rest})
\end{aligned}$$

extending the symmetry sum to $\gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\}$. As always, we used harmonics decomposition. The following chain, similar to Eq. (A.58), holds:

$$\begin{aligned}
w^{(\gamma)} &\equiv \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} w_{\mathbf{k}} \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \langle \hat{\phi}_{\mathbf{k}}^\dagger \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \langle \hat{\Phi}_{\mathbf{k}}^\dagger \tau^+ \hat{\Phi}_{\mathbf{k}} \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \left[\langle \hat{s}_{\mathbf{k}}^x \rangle + i \langle \hat{s}_{\mathbf{k}}^y \rangle\right] \\
&= -\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta]
\end{aligned} \tag{2.23}$$

where $\tau^+ \equiv \tau^x + i\tau^y$. We used Eqns. (2.5), (2.6) and (2.8) as well as the thermal factor defined in Eq. (2.9). The derivation above holds for all symmetries, including the s one as testifies Eq. (A.58). Explicitly, five self-consitency equations are then derived,

$$w^{(s^*)} = \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\text{MBZ}} (\cos k_x + \cos k_y) \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \tag{2.24}$$

$$w^{(d)} = \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\text{MBZ}} (\cos k_x - \cos k_y) \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \tag{2.25}$$

$$-iw^{(p_x)} = \frac{1}{\sqrt{2} L_x L_y} \sum_{\mathbf{k}\in\text{MBZ}} \sin k_x \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \tag{2.26}$$

$$-iw^{(p_y)} = \frac{1}{\sqrt{2} L_x L_y} \sum_{\mathbf{k}\in\text{MBZ}} \sin k_y \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \tag{2.27}$$

Of these, the singlet channel admits for non-zero fluctuations of the superconducting order parameter just in the s^* and d channels. This is because the gap function is inversion-symmetric and p_ℓ channels equations integrate it with an antisymmetric function.

Finally, the renormalized s-wave part, originated by the Cooper channel of $\hat{H}_U$ and discussed for the HM in Eq. (A.41) is given by:

$$w^{(s)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\text{BZ}} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \tag{2.28}$$

Let us now move to the computation of the contraction energy.

2.5.2 Cooper contraction energy

The Cooper contraction energy is immediately derived from Eq. (2.22),

$$\begin{aligned} E^{(\mathrm{SC})}_{\mathrm{C}/V/\mathrm{o.s.}} &= -L_x L_y \times V \sum_\gamma \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} w_{\mathbf{q}} \right] \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}'\in\mathrm{BZ}} \varphi^{(\gamma)+}_{\mathbf{q}'} w^*_{\mathbf{q}'} \right] \\ &= -L_x L_y \times \sum_\gamma \left| w^{(\gamma)} \right|^2 \end{aligned}$$

and is completed by its local counterpart of Eq.

2.6 Full MFT hamiltonian and anisotropic pairing

By composing all three channels we are able to derive the final form of the MFT gran-canonical hamiltonian,

$$\begin{aligned} \hat{H} - \mu \hat{N} \simeq \sum_{\mathbf{k}\in\mathrm{MBZ}} \tilde{\xi}_{\mathbf{k}\sigma} \left[\hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}_{\mathbf{k}\uparrow} + \hat{c}_{-\mathbf{k}\downarrow} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \right] - \sum_\sigma \sum_{\mathbf{k}\in\mathrm{BZ}} \left[\tilde{\Delta}_{\mathbf{k}} \hat{\phi}_{\mathbf{k}} + \tilde{\Delta}^*_{\mathbf{k}} \hat{\phi}^\dagger_{\mathbf{k}} \right] \\ - \left[E^{(\mathrm{SC})}_{\mathrm{A}/t} + E^{(\mathrm{SC})}_{\mathrm{H}/U} + E^{(\mathrm{SC})}_{\mathrm{C}/U} + E^{(\mathrm{SC})}_{\mathrm{H}/V} + E^{(\mathrm{SC})}_{\mathrm{F}/V} + E^{(\mathrm{SC})}_{\mathrm{C}/U} \right] \end{aligned}$$

having we included the "anticommutation energy" also defined for the conventional HM in Eq. (A.45),

$$E^{(\mathrm{SC})}_{\mathrm{A}/t} \equiv - \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{\xi}_{\mathbf{k}}$$

coming from fermionic anticommutation relations that are employed in order to get the kinetic term. The renormalized bands are

$$\tilde{\xi}_{\mathbf{k}\sigma} = \epsilon_{\mathbf{k}\sigma} + V \sum_{\gamma\in\{s^*,d\}} u^{(\gamma)} \varphi^{(\gamma)*}_{\mathbf{k}} - \tilde{\mu}$$

and the full gap function by

$$\tilde{\Delta}_{\mathbf{k}\sigma} = -U w^{(s)} + V \sum_\gamma w^{(\gamma)} \varphi^{(\gamma)*}_{\mathbf{k}}$$

We can identify the gap harmonics by

$$\Delta^{(s)} = -U w^{(s)} \qquad \Delta^{(\gamma)} = V w^{(\gamma)} \quad \text{being} \quad \gamma \in \{s^*, p_x, p_y, d\}$$

Define the full gap function as the sum of both contributions,

$$\Delta_{\mathbf{k}} \equiv \mathcal{V}_{\mathbf{k}} - \mathcal{U}_{\mathbf{k}} \tag{2.29}$$

The full self-consistency equation is given by the simple combination of Eqns. (**??**) and (**??**),

$$\Delta_{\mathbf{k}} \equiv \sum_{\mathbf{k}'} \left[V^{(s)} + V_{\mathbf{k}\mathbf{k}'} \right] \langle \hat{\phi}^\dagger_{\mathbf{k}'} \rangle \quad \text{with} \quad V^{(s)} = -\frac{U}{2 L_x L_y} \tag{2.30}$$

The gap function decomposes in symmetry channels as well,

$$\Delta_{\mathbf{k}} = \sum_\gamma \Delta^{(\gamma)} \varphi^{(\gamma)}_{\mathbf{k}}$$

Structure	Self-consistency equation	Graph
s-wave	$\Delta^{(s)} = -\frac{U}{2L_xL_y}\sum_{\mathbf{k}}\langle\hat{\phi}^\dagger_{\mathbf{k}}\rangle$	Fig. ??
Extended s-wave	$\Delta^{(s^*)} = \frac{V}{L_xL_y}\sum_{\mathbf{k}}(c_x+c_y)\langle\hat{\phi}^\dagger_{\mathbf{k}}\rangle$	Fig. ??
p_x-wave	$\Delta^{(p_x)} = -i\sqrt{2}\frac{V}{L_xL_y}\sum_{\mathbf{k}}s_x\langle\hat{\phi}^\dagger_{\mathbf{k}}\rangle$	Fig. ??
p_y-wave	$\Delta^{(p_y)} = -i\sqrt{2}\frac{V}{L_xL_y}\sum_{\mathbf{k}}s_y\langle\hat{\phi}^\dagger_{\mathbf{k}}\rangle$	Fig. ??
$d_{x^2-y^2}$-wave	$\Delta^{(d)} = \frac{V}{L_xL_y}\sum_{\mathbf{k}}(c_x-c_y)\langle\hat{\phi}^\dagger_{\mathbf{k}}\rangle$	Fig. ??

Table 2.3 Symmetry resolved self-consistency equations for the MFT parameters $\Delta^{(\gamma)}$, based on Eq. (2.33) and (2.34). By computing $\langle\hat{\phi}^\dagger_{\mathbf{k}}\rangle$, it is possible to reconstruct the various components of the gap function.

If SC arises in a specific symmetry channel, $\Delta_{\mathbf{k}}$ will show the same symmetry. It follows, due to orthonormality and using Eq. (2.33),

$$\begin{aligned}\Delta^{(\gamma)} &= \frac{1}{L_xL_y}\sum_{\mathbf{k}}\varphi^{(\gamma)*}_{\mathbf{k}}\Delta_{\mathbf{k}}\\ &= \frac{1}{L_xL_y}\sum_{\mathbf{k}}\varphi^{(\gamma)*}_{\mathbf{k}}\sum_{\mathbf{k}'}\left[V^{(s)}+V_{\mathbf{k}\mathbf{k}'}\right]\langle\hat{\phi}^\dagger_{\mathbf{k}'}\rangle\\ &= \frac{1}{L_xL_y}\sum_{\mathbf{k}}\varphi^{(\gamma)*}_{\mathbf{k}}\sum_{\mathbf{k}'\gamma'}V^{(\gamma')}\varphi^{(\gamma')}_{\mathbf{k}}\varphi^{(\gamma')*}_{\mathbf{k}'}\langle\hat{\phi}^\dagger_{\mathbf{k}'}\rangle\\ &= V^{(\gamma)}\sum_{\mathbf{k}}\varphi^{(\gamma)*}_{\mathbf{k}}\langle\hat{\phi}^\dagger_{\mathbf{k}}\rangle\end{aligned} \tag{2.31}$$

This result provides a set of self-consistency equations for each symmetry channel, listed in Tab. 2.3. Notice that to reconstruct self-consistently the full s-wave phase transition, the actual gap function is given by

$$\Delta^{(s)} + \Delta^{(s^*)}(c_x+c_y)$$

The s-wave transition is the only one equipped of both the local and the non-local parts.

2.7 SU free energy density

BCS essentially relies on the insurgence of an effective free fermionic field, described by the Bogoliubov bands, and the condensation of Cooper pairs. Superconducting state is favored by the lowering of free energy granted by the shifting of much of the spectral weight to a lower-energy maximum [**zhou1992density**]. "Condensation energy" is simply the energy difference between the superconducting state and the normal state, and its negativity indicates instability towards superconducting order. It's pretty straightforward to derive the ground state energy for the standard BCS state. When performing Wick's contractions,

$$\begin{aligned}\langle\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\hat{c}_\gamma\hat{c}_\delta\rangle \sim\ & \langle\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\rangle\hat{c}_\gamma\hat{c}_\delta + \hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\langle\hat{c}_\gamma\hat{c}_\delta\rangle - \langle\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\rangle\langle\hat{c}_\gamma\hat{c}_\delta\rangle && \text{Cooper terms}\\ & - \langle\hat{c}^\dagger_\alpha\hat{c}_\gamma\rangle\hat{c}^\dagger_\beta\hat{c}_\delta - \hat{c}^\dagger_\alpha\hat{c}_\gamma\langle\hat{c}^\dagger_\beta\hat{c}_\delta\rangle + \langle\hat{c}^\dagger_\alpha\hat{c}_\gamma\rangle\langle\hat{c}^\dagger_\beta\hat{c}_\delta\rangle && \text{Fock terms}\\ & + \langle\hat{c}^\dagger_\alpha\hat{c}_\delta\rangle\hat{c}^\dagger_\beta\hat{c}_\gamma + \hat{c}^\dagger_\alpha\hat{c}_\delta\langle\hat{c}^\dagger_\beta\hat{c}_\gamma\rangle - \langle\hat{c}^\dagger_\alpha\hat{c}_\delta\rangle\langle\hat{c}^\dagger_\beta\hat{c}_\gamma\rangle && \text{Hartree terms}\end{aligned}$$

Form each of the three rows above, the last term is the "leftover" part which shifts energy. Thus it can be ignored in derivation, but becomes important when considering the energy in the end.

The EHM hamiltonian can be written as:

$$\hat{H} = \sum_{\mathbf{k}} \xi_{\mathbf{k}} \left[\hat{c}^\dagger_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} + \mathbb{I} - \hat{c}_{-\mathbf{k}\downarrow}\hat{c}^\dagger_{-\mathbf{k}\downarrow}\right] - \sum_{\mathbf{k},\mathbf{k}'} \left[V^{(s)} + V_{\mathbf{k}\mathbf{k}'}\right] \hat{\phi}^\dagger_{\mathbf{k}}\hat{\phi}_{\mathbf{k}'} - \frac{2V}{L_x L_y} \sum_{\mathbf{k},\mathbf{k}'} \sum_{\sigma} \left[\cos(\delta k_x) + \cos(\delta k_y)\right] \hat{n}_{\mathbf{k}\sigma}\hat{n}_{\mathbf{k}'\sigma} \tag{2.32}$$

as in Eq. (2.33). Following the scheme above, the "leftovers" of the MFT hamiltonian are just

$$\sum_{\mathbf{k},\mathbf{k}'} \Delta_{\mathbf{k}} \langle \hat{\phi}_{\mathbf{k}'} \rangle - \frac{2V}{L_x L_y} \sum_{\mathbf{k},\mathbf{k}'} \sum_{\sigma} \left[\cos(\delta k_x) + \cos(\delta k_y)\right] g_{\mathbf{k}} g_{\mathbf{k}'}$$

Further developing this result and applying Eqns. (**??**), (**??**), these last terms can be written as:

$$\sum_{\mathbf{k}} \frac{|\Delta_{\mathbf{k}}|^2}{2\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) - 2V L_x L_y \sum_{\gamma} \left|g^{(\gamma)}\right|^2 \qquad \gamma \in \{s^*, d_{x^2-y^2}\} \tag{2.33}$$

as for Eq. (**??**). Notice the presence of a volume factor in the $\mathcal{O}\left(g^2\right)$ correction.

Now, we retrieve the expression for the MFT free energy density [**hutchinson2021mixed**] simply by adding four contributions:

1. Bogoliubov free quasiparticles thermal ground state energy density,

$$\frac{1}{L_x L_y} \sum_{\mathbf{k}} \tilde{E}_{\mathbf{k}} \left[f(\tilde{E}_{\mathbf{k}};\beta,0) - f(-\tilde{E}_{\mathbf{k}};\beta,0)\right] = \frac{1}{L_x L_y} \sum_{\mathbf{k}} \tilde{E}_{\mathbf{k}} \left[f(\tilde{E}_{\mathbf{k}};\beta,0) - \left(1 - f(\tilde{E}_{\mathbf{k}};\beta,0)\right)\right]$$
$$= \frac{2}{L_x L_y} \sum_{\mathbf{k}} \tilde{E}_{\mathbf{k}} f(\tilde{E}_{\mathbf{k}};\beta,0) - \frac{1}{L_x L_y} \sum_{\mathbf{k}} \tilde{E}_{\mathbf{k}}$$

   having used the simple property of the Fermi-Dirac distribution $f(-x;\beta,0) = 1 - f(-x;\beta,0)$.

2. The term originated by the identity produced by fermionic anticommutation rules in Eq. (2.35),

$$\frac{1}{L_x L_y} \sum_{\mathbf{k}} \tilde{\xi}_{\mathbf{k}}$$

   averaged on the lattice.

3. The "leftovers" of Eq. (2.36), divided by $L_x L_y$.

4. The entropic free energy

$$-Ts$$

   with $T = 1/k_B\beta$ and s the free Fermi gas entropy density for the Bogoliubov quasiparticles,

$$s = -\frac{2k_B}{L_x L_y} \sum_{\mathbf{k}} \left[\left(1 - f(\tilde{E}_{\mathbf{k}};\beta,0)\right) \ln\left(1 - f(\tilde{E}_{\mathbf{k}};\beta,0)\right) + f(\tilde{E}_{\mathbf{k}};\beta,0) \ln f(\tilde{E}_{\mathbf{k}};\beta,0)\right]$$

   We use the identity:

$$\ln\left(1 - \frac{1}{e^x + 1}\right) = x + \ln\left(\frac{1}{e^x + 1}\right)$$

   and reduce the above expression to:

$$s = -\frac{2k_B}{L_x L_y} \sum_{\mathbf{k}} \left[\ln\left(1 - f(\tilde{E}_{\mathbf{k}};\beta,0)\right) + f(\tilde{E}_{\mathbf{k}};\beta,0) \ln f(\tilde{E}_{\mathbf{k}};\beta,0) - \beta\tilde{E}_{\mathbf{k}} f(\tilde{E}_{\mathbf{k}};\beta,0) - f(\tilde{E}_{\mathbf{k}};\beta,0) \ln f(\tilde{E}_{\mathbf{k}};\beta,0)\right]$$

   Second and fourth terms cancel out, while the third compensates the first one of contribution 1.

All together, the MFT averaged free energy looks like this:

$$f_{\mathrm{MFT}}[\Delta_{\mathbf{k}}, g_{\mathbf{k}}] = \frac{1}{L_x L_y} \sum_{\mathbf{k}} \left[\tilde{\xi}_{\mathbf{k}} - \tilde{E}_{\mathbf{k}} + \frac{|\Delta_{\mathbf{k}}|^2}{2\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2} \right) \right] - 2V \sum_{\gamma \in \{s^*, d\}} \left| g^{(\gamma)} \right|^2 + \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}} \ln\left(1 - f(\tilde{E}_{\mathbf{k}}; \beta, 0) \right) \tag{2.34}$$

The two main differences with the commonly known expressions for f_{MFT} [**hutchinson2021mixed**, **senarath2025mixed**] are the presence of bands renormalization , the absence of $-\mu n$ [Why?], and the $\mathcal{O}\left(g^2\right)$ correction. The latter, not only is negative for attractive V, a feature that can lead to superconducting phase energy lowering and stabilisation, but also it depends explicitly on the spectral amplitude of the s^* and d-wave components of $g_{\mathbf{k}}$. Since as seen the d-wave part of $g_{\mathbf{k}}$ arises for mixed symmetries, this details seems to indicate an energetic preference for multi-symmetry phases.

2.8 Numerical simulations

[To be continued. . .]

2.8.1 HF iterative algorithm sketch

[To be continued. . .]

Appendix A

Mean-Field Theory in Hubbard lattices

In this Appendix the MFT solutions to the conventional Hubbard hamiltonian of Eq. (**??**),

$$\hat{H} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \left[\hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + \text{h.c.} \right] + U \sum_{i} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} \qquad t, U > 0 \tag{A.1}$$

are described. The discussion is limited to the two-dimensional square lattice. The results of ths appendix are relevant and used throughout the text as the groundwork to be extended by the means of MFT applied to the non-local part of the EHM.

A.1 Antiferromagnetic phase

We work in a commensurate AF phase ordering, which is, we explicitly break symmetry by realizing a SDW phase with spin density unevenly distributed on sites with periodicity of two sites, each direction. Recalling the discussion of Sec. **??**, AF long-range ordering we deal with here lies within the $\mathrm{SU}(2) \to \mathrm{U}^z(1)$ SSB, represented pictorially in Fig. **??**. Recalling Tab. **??**, for such a physical phase the only Wick contraction channel available is the Hartree one. Let us proceed in such sense.

A.1.1 MFT treatment of the Hartree channel

Let us change notation, indicating each site as

$$i \to \mathbf{r} = (x, y) \qquad x, y \in \mathbb{N}$$

In terms of the physical magnetization m, the MFT Ansatz is given by

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = n - (-1)^{x+y+\delta_{\sigma=\uparrow}} m \tag{A.2}$$

Decoupling the HM hamiltonian and substituting,

$$\begin{aligned} \hat{H} &\simeq -t \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \left[\hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + \text{h.c.} \right] + U \sum_{\mathbf{r}} \left[\hat{n}_{\mathbf{r}\uparrow} \langle \hat{n}_{\mathbf{r}\downarrow} \rangle + \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \hat{n}_{\mathbf{r}\downarrow} - \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \langle \hat{n}_{\mathbf{r}\downarrow} \rangle \right] \\ &= -t \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \left[\hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + \text{h.c.} \right] + nU \sum_{\mathbf{r}} \left[\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow} \right] - mU \sum_{\mathbf{r}} (-1)^{x+y} \left[\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow} \right] - E^{(\mathrm{AF})}_{\mathrm{H}/U} \end{aligned} \tag{A.3}$$

with $E^{(\mathrm{AF})}_{\mathrm{H}/U}$ the HM Hartree contraction energy,

$$\begin{aligned} E^{(\mathrm{AF})}_{\mathrm{H}/U} &\equiv U \sum_{\mathbf{r}} \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \langle \hat{n}_{\mathbf{r}\downarrow} \rangle \\ &= U \sum_{\mathbf{r}} \left(n + (-1)^{x+y} m \right) \left(n - (-1)^{x+y} m \right) \\ &= L_x L_y \times U \left(n^2 - m^2 \right) \end{aligned} \tag{A.4}$$

Figure A.1 | Depiction of the Hubbard square lattice hopping band $\epsilon_{\mathbf{k}} = -2t[\cos(k_x) + \cos(k_y)]$. The red lines mark the zero-energy intersection.

Figure A.2 | Alternative depiction of the Hubbard square lattice hopping band previously reported in Fig. A.1. The Magnetic Brillouin Zone (MBZ) is delimited by the zero-energy contour and is indicated in figure. As it is evident, energy sign flips by taking a (π, π) translation in $\mathbf{k}$ space.

By a Fourier transform, the phase factor of Eq. (A.3) can be absorbed in the destruction operator inside of $\hat{n}_{\mathbf{r}\sigma}$:

$$\begin{aligned}
\sum_{\mathbf{r}} (-1)^{x+y} \hat{n}_{\mathbf{r}\sigma} &= \sum_{\mathbf{r}} (-1)^{x+y} \hat{c}^\dagger_{\mathbf{r}\sigma} \hat{c}_{\mathbf{r}\sigma} \\
&\overset{\mathcal{F}}{=} \sum_{\mathbf{r}} e^{i\boldsymbol{\pi}\cdot\mathbf{r}} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}\in\mathrm{BZ}} e^{i\mathbf{k}\cdot\mathbf{r}} \hat{c}^\dagger_{\mathbf{k}\sigma} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}'\in\mathrm{BZ}} e^{-i\mathbf{k}'\cdot\mathbf{r}} \hat{c}_{\mathbf{k}'\sigma} \\
&= \sum_{\mathbf{k}\in\mathrm{BZ}} \sum_{\mathbf{k}'\in\mathrm{BZ}} \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}'\sigma} \frac{1}{L_x L_y} \sum_{\mathbf{r}} e^{-i[\mathbf{k}'-(\mathbf{k}+\boldsymbol{\pi})]\cdot\mathbf{r}} \\
&= \sum_{\mathbf{k}\in\mathrm{BZ}} \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}
\end{aligned} \tag{A.5}$$

where $\boldsymbol{\pi} = (\pi, \pi)$. It follows:

$$mU \sum_{\mathbf{r}} (-1)^{x+y} \left[\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}\right] = mU \sum_{\mathbf{k}\in\mathrm{BZ}} \left[\hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\uparrow} - \hat{c}^\dagger_{\mathbf{k}\downarrow} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\downarrow}\right]$$

Note that the Hartree channel renormalizes chemical potential,

$$nU \sum_{\mathbf{r}} \left[\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}\right] = nU \times \hat{N} \qquad \Longrightarrow \qquad \mu \to \tilde{\mu} \equiv \mu - nU$$

an effect that is present also in the EHM.

A.1.2 Nambu representation of the AF phase and diagonalization

Consider the band of Fig. A.1 . As does **fabrizio2022course** [**fabrizio2022course**], the area delimited externally by the solid lines at zero energy is denominated "Magnetic Brillouin Zone" (MBZ). The periodicity of $\mathbf{k}$ space guarantees that the full BZ can be taken as well to be the one of Fig. **??**. Then:

$$\begin{aligned}
\sum_{\mathbf{k}\in\mathrm{BZ}} \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} &= \sum_{\mathbf{k}\in\mathrm{MBZ}} \left[\hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} + \hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma} \hat{c}_{\mathbf{k}+2\boldsymbol{\pi}\sigma}\right] \\
&= \sum_{\mathbf{k}\in\mathrm{MBZ}} \left[\hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} + \hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma} \hat{c}_{\mathbf{k}\sigma}\right]
\end{aligned} \tag{A.6}$$

Periodicity by shifts $2\boldsymbol{\pi}$ has been used. Now, define the Nambu spinors:

$$\hat{\Psi}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{bmatrix}$$

As in Eq. [Missing], it is useful to define the density and AF fluctuation operators:

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \tag{A.7}$$

$$\hat{\psi}_{\mathbf{k}\sigma} \equiv \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \tag{A.8}$$

whose EVs are relevant order parameters for the normal and AF phases. Let also the gap function:

$$\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU$$

Figure A.3 | Pseudo-magnetic field originating from mean-field treatment of the square Hubbard hamiltonian. Here, only the $\sigma =\uparrow$ situation is plotted.

At fixed filling, the nU term is a pure chemical potential shift, thus will be neglected. The kinetic term transforms as

$$\begin{aligned}
-t\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\left[\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}+\text{h.c.}\right] &\overset{\mathcal{F}}{=} -t\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\frac{1}{\sqrt{L_xL_y}}\sum_{\mathbf{k}\in\text{BZ}}e^{i\mathbf{k}\cdot\mathbf{r}}\hat{c}^{\dagger}_{\mathbf{k}\sigma}\frac{1}{\sqrt{L_xL_y}}\sum_{\mathbf{k}'\in\text{BZ}}e^{-i\mathbf{k}'\cdot\mathbf{r}'}\hat{c}_{\mathbf{k}'\sigma}+\text{h.c.}\\
&= -2t\sum_{\mathbf{k},\mathbf{k}'}\sum_{\sigma}\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma}\frac{1}{L_xL_y}\sum_{\mathbf{r}\in\mathcal{S}/2}e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}}\left[e^{ik'_x}+e^{-ik'_x}+e^{ik'_y}+e^{-ik'_y}\right]\\
&= \sum_{\mathbf{k}\in\text{BZ}}\sum_{\sigma}\epsilon_{\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}\\
&= \sum_{\mathbf{k}\in\text{MBZ}}\sum_{\sigma}\left[\epsilon_{\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}+\epsilon_{\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}^{\dagger}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right]\\
&= \sum_{\mathbf{k}\in\text{MBZ}}\sum_{\sigma}\epsilon_{\mathbf{k}\sigma}\left[\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}-\hat{c}^{\dagger}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right]
\end{aligned}$$

where $\epsilon_{\mathbf{k}\sigma}\equiv -2t(\cos k_x+\cos k_y)$. In the fourth passage, the sum over the full BZ was written considering that the entirety of the zone is given by all the points in the MBZ plus their conjugates obtained by a $\boldsymbol{\pi}$ shift in the flipped band. As depicted in Fig. **??**, kinetic energy is anti-periodic in $\mathbf{k}$ space by a vector $\boldsymbol{\pi}$. This anti-periodicity accounts for the minus sign arising in the last passage. The hamiltonian is then given by:

$$\hat{H}\simeq\sum_{\mathbf{k}\in\text{MBZ}}\sum_{\sigma}\hat{\Psi}^{\dagger}_{\mathbf{k}\sigma}h_{\mathbf{k}\sigma}\hat{\Psi}_{\mathbf{k}\sigma}-E^{(\text{AF})}_{\text{H}/U}\quad\text{being}\quad h_{\mathbf{k}\sigma}\equiv\begin{bmatrix}\epsilon_{\mathbf{k}\sigma} & -\Delta_{\mathbf{k}\sigma}\\ -\Delta^{*}_{\mathbf{k}\sigma} & -\epsilon_{\mathbf{k}\sigma}\end{bmatrix}\tag{A.9}$$

Notice: the Nambu hamiltonian is a 2×2 matrix over the MBZ – which is half the full BZ, coherently with a solution which essentially bipartites the lattice giving back a double sized unit cell.

The system ground-state is obtained by the means of a Bogoliubov rotation. The hamiltonian maps onto the simple one of a spin in a magnetic field,

$$h_{\mathbf{k}\sigma}=\epsilon_{\mathbf{k}\sigma}\tau^z-\Delta_{\mathbf{k}\sigma}\tau^x$$

being τ^{α} the Pauli matrices. Then, defining

$$\hat{s}^{\alpha}_{\mathbf{k}\sigma}\equiv\hat{\Psi}^{\dagger}_{\mathbf{k}\sigma}\tau^{\alpha}\hat{\Psi}_{\mathbf{k}\sigma}\quad\text{and}\quad\mathbf{b}_{\mathbf{k}\sigma}\equiv\begin{bmatrix}-\Delta_{\mathbf{k}\sigma}\\ 0\\ \epsilon_{\mathbf{k}\sigma}\end{bmatrix}$$

one gets:

$$\hat{H}\simeq\sum_{\mathbf{k}\in\text{MBZ}}\sum_{\sigma}\mathbf{b}_{\mathbf{k}\sigma}\cdot\hat{\mathbf{s}}_{\mathbf{k}\sigma}-E^{(\text{AF})}_{\text{H}/U}\quad\text{where}\quad\hat{\mathbf{s}}_{\mathbf{k}\sigma}=\begin{bmatrix}\hat{s}^{x}_{\mathbf{k}\sigma}\\ \hat{s}^{y}_{\mathbf{k}\sigma}\\ \hat{s}^{z}_{\mathbf{k}\sigma}\end{bmatrix}\tag{A.10}$$

The hamiltonian represents a system of spins subject to local magnetic fields, each tilted by an angle $\tan(2\theta_{\mathbf{k}\sigma})=\Delta_{\mathbf{k}\sigma}/\epsilon_{\mathbf{k}\sigma}$, as sketched in Fig. A.3. Evidently the following relations, also reported in Tab. 1.1, hold:

$$\hat{s}^{x}_{\mathbf{k}\sigma}=\hat{\psi}_{\mathbf{k}\sigma}+\hat{\psi}^{\dagger}_{\mathbf{k}\sigma}\tag{A.11}$$

$$\hat{s}^{y}_{\mathbf{k}\sigma}=-i\left(\hat{\psi}_{\mathbf{k}\sigma}-\hat{\psi}^{\dagger}_{\mathbf{k}\sigma}\right)\tag{A.12}$$

$$\hat{s}^{z}_{\mathbf{k}\sigma}=\hat{n}_{\mathbf{k}\sigma}-\hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\tag{A.13}$$

Let also:

$$\begin{aligned}\hat{\mathbb{I}}_{\mathbf{k}\sigma}&\equiv\hat{\Psi}^{\dagger}_{\mathbf{k}\sigma}\mathbb{I}_{2\times 2}\hat{\Psi}_{\mathbf{k}\sigma}\\ &=\hat{n}_{\mathbf{k}\sigma}+\hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\end{aligned}\tag{A.14}$$

Figure A.4 | Depiction of the AF bands $\pm E_{\mathbf{k}}$. [To be continued...]

which accounts for a chemical potential shift.

Diagonalization of each $h_{\mathbf{k}\sigma}$ is obtained trivially by a simple rotation around the y axis:

$$d_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma} h_{\mathbf{k}\sigma} W^\dagger_{\mathbf{k}\sigma} \quad \text{with} \quad d_{\mathbf{k}\sigma} \equiv \begin{bmatrix} E_{\mathbf{k}\sigma} & \\ & -E_{\mathbf{k}\sigma} \end{bmatrix}$$

and where the diagonalization unitary matrix is simply a rotation by an angle $2\theta_{\mathbf{k}\sigma}$ around the y axis, counter-clockwise

$$\begin{aligned} W_{\mathbf{k}\sigma} &\equiv e^{-i\theta_{\mathbf{k}\sigma}\tau^y} \\ &= \cos\theta_{\mathbf{k}\sigma} - i\sin\theta_{\mathbf{k}\sigma}\tau^y \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}\sigma} & -\sin\theta_{\mathbf{k}\sigma} \\ \sin\theta_{\mathbf{k}\sigma} & \cos\theta_{\mathbf{k}\sigma} \end{bmatrix} \end{aligned} \tag{A.15}$$

Eigenvalues are:

$$E_{\mathbf{k}\sigma} \equiv \sqrt{\epsilon^2_{\mathbf{k}\sigma} + |\Delta_{\mathbf{k}\sigma}|^2} \tag{A.16}$$

At any finite temperature, the ground-state of such a system is well-known: a thermal superposition of "down" spins with "up" spins. In the pseudospin picture, at each "site" ($\mathbf{k}\sigma$), the lowest-energy state is the one anti-aligned with the pseudo-magnetic field. It is useful now to define the thermal factors,

$$\mathrm{Th}^{(\pm)}_{\mathbf{k}\sigma}[\beta,\mu] \equiv f\left(E_{\mathbf{k}\sigma};\beta,\mu\right) \pm f\left(-E_{\mathbf{k}\sigma};\beta,\mu\right) \tag{A.17}$$

that will appear quite often. Then it's immediate to derive the various expectation values.

$$\langle \hat{s}^x_{\mathbf{k}\sigma}\rangle = -\frac{\mathrm{Re}\{\Delta_{\mathbf{k}\sigma}\}}{2E_{\mathbf{k}\sigma}}\mathrm{Th}^{(-)}_{\mathbf{k}\sigma}[\beta,\mu] \tag{A.18}$$

$$\langle \hat{s}^y_{\mathbf{k}\sigma}\rangle = -\frac{\mathrm{Im}\{\Delta_{\mathbf{k}\sigma}\}}{2E_{\mathbf{k}\sigma}}\mathrm{Th}^{(-)}_{\mathbf{k}\sigma}[\beta,\mu] \tag{A.19}$$

$$\langle \hat{s}^z_{\mathbf{k}\sigma}\rangle = \frac{\epsilon_{\mathbf{k}\sigma}}{2E_{\mathbf{k}\sigma}}\mathrm{Th}^{(-)}_{\mathbf{k}\sigma}[\beta,\mu] \tag{A.20}$$

Note that considering the gran canonical hamiltonian $\hat{K} \equiv \hat{H} - \mu\hat{N}$ and using Eq. (A.14) the diagonalizing matrix remains the same, being

$$-\mu\hat{N} = \sum_{\mathbf{k}\in\mathrm{MBZ}} \begin{bmatrix} -\mu & \\ & -\mu \end{bmatrix} \quad \Longrightarrow \quad W_{\mathbf{k}\sigma}\left[h_{\mathbf{k}\sigma} - \mu\mathbb{I}_{2\times2}\right] W^\dagger_{\mathbf{k}\sigma} = d_{\mathbf{k}\sigma} - \mu\mathbb{I}_{2\times2} \tag{A.21}$$

thus for non-null chemical potential (i.e. finite doping) the eigenvalues are just shifted, $\pm E_{\mathbf{k}} - \mu$. In other words, the new bands retain the same chemical potential of the bare bands. Notice also that the presence of an absolute value makes the eigenvalues independent of the σ index. Eigenvector fermion operators are obtained simply as:

$$\hat{\Gamma}_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma}\hat{\Psi}_{\mathbf{k}\sigma} \tag{A.22}$$

The $\hat{\Gamma}$ spinor operators are effectively free fermionic spinor fields and as such must be treated. Explicitly,

$$\begin{aligned} \hat{\Gamma}_{\mathbf{k}\sigma} &= \begin{bmatrix} \cos\theta_{\mathbf{k}\sigma} & -\sin\theta_{\mathbf{k}\sigma} \\ \sin\theta_{\mathbf{k}\sigma} & \cos\theta_{\mathbf{k}\sigma} \end{bmatrix} \begin{bmatrix} \hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} - \sin\theta_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \\ \sin\theta_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} + \cos\theta_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{bmatrix} \equiv \begin{bmatrix} \hat{\gamma}^{(+)}_{\mathbf{k}\sigma} \\ \hat{\gamma}^{(-)}_{\mathbf{k}\sigma} \end{bmatrix} \end{aligned} \tag{A.23}$$

The consequence of Eq. (A.21) is interesting by itself: within this HF solution, the gap opens always at $E = 0$. In other words, the bands sketched in Fig. A.4 retain the same geometric features

regardless of the doping, what changes is the bands width and the gap between them. The very key point here is that, being the two bands symmetric (thus containing each half of the electronic states) and defined over the MBZ, any finite doping $\delta > 0$ will produce a metallic behaviour, imposing that the chemical potential crosses the upper band. It is widely observed and confirmed by more refined numerical procedures, however, that AF establishes an insulating phase – here only reproduced at half-filling, where the chemical potential lies within the gap. As will be detailed in A.1.7, the commensurate nature of AF SSB at wavevector $\boldsymbol{\pi}$ of this self-consistent solution is highly unstable and atypical at finite doping [**singh1990collective**].

A.1.3 Explicit zero-temperature AF ground state at half-filling

The AF ground state can be expressed easily at $n = 1/2$ and $\beta \to +\infty$. Let $|\Uparrow_{\mathbf{k}\sigma}\rangle$ and $|\Downarrow_{\mathbf{k}\sigma}\rangle$ be respectively the "up" and "down" $\mathbf{k}$-th pseudospin states. Let $\left|\overline{\Downarrow}_{\mathbf{k}\sigma}\right\rangle$ be the "down" state in the tilted frame. The ground state is given by

$$|\Psi\rangle \equiv \bigotimes_{\sigma} \bigotimes_{\mathbf{k}\in\mathrm{MBZ}} \left|\overline{\Downarrow}_{\mathbf{k}\sigma}\right\rangle \tag{A.24}$$

We can link the "down" states via an inverse rotation,

$$\begin{aligned}
\left|\overline{\Downarrow}_{\mathbf{k}\sigma}\right\rangle &= e^{-i\theta_{\mathbf{k}\sigma}\hat{s}^y_{\mathbf{k}\sigma}} |\Downarrow_{\mathbf{k}\sigma}\rangle \\
&= \cos\theta_{\mathbf{k}\sigma} |\Downarrow_{\mathbf{k}\sigma}\rangle - i\sin\theta_{\mathbf{k}\sigma}\hat{s}^y_{\mathbf{k}\sigma} |\Downarrow_{\mathbf{k}\sigma}\rangle \\
&= \cos\theta_{\mathbf{k}\sigma} |\Downarrow_{\mathbf{k}\sigma}\rangle - \sin\theta_{\mathbf{k}\sigma}(\hat{s}^+_{\mathbf{k}\sigma} - \hat{s}^-_{\mathbf{k}\sigma}) |\Downarrow_{\mathbf{k}\sigma}\rangle \\
&= \cos\theta_{\mathbf{k}\sigma} |\Downarrow_{\mathbf{k}\sigma}\rangle - \sin\theta_{\mathbf{k}\sigma} |\Uparrow_{\mathbf{k}\sigma}\rangle
\end{aligned} \tag{A.25}$$

having we used

$$\hat{s}^{\pm}_{\mathbf{k}\sigma} \equiv \frac{\hat{s}^x_{\mathbf{k}\sigma} \pm i\hat{s}^y_{\mathbf{k}\sigma}}{2}$$

Use now Eqns. (A.14) and (A.13),

$$\frac{\hat{\mathbb{I}}_{\mathbf{k}\sigma} + \hat{s}^z_{\mathbf{k}\sigma}}{2} = \hat{n}_{\mathbf{k}\sigma} \qquad \frac{\hat{\mathbb{I}}_{\mathbf{k}\sigma} - \hat{s}^z_{\mathbf{k}\sigma}}{2} = \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma}$$

Evaluating the above operators on the "up" and "down" states, we get immediately

$$\begin{aligned}
&\frac{1}{2}\langle\Uparrow_{\mathbf{k}\sigma}|\hat{\mathbb{I}}_{\mathbf{k}\sigma} + \hat{s}^z_{\mathbf{k}\sigma}|\Uparrow_{\mathbf{k}\sigma}\rangle = 1 \qquad &&\frac{1}{2}\langle\Uparrow_{\mathbf{k}\sigma}|\hat{\mathbb{I}}_{\mathbf{k}\sigma} - \hat{s}^z_{\mathbf{k}\sigma}|\Uparrow_{\mathbf{k}\sigma}\rangle = 0 \\
&\frac{1}{2}\langle\Downarrow_{\mathbf{k}\sigma}|\hat{\mathbb{I}}_{\mathbf{k}\sigma} + \hat{s}^z_{\mathbf{k}\sigma}|\Downarrow_{\mathbf{k}\sigma}\rangle = 0 \qquad &&\frac{1}{2}\langle\Downarrow_{\mathbf{k}\sigma}|\hat{\mathbb{I}}_{\mathbf{k}\sigma} - \hat{s}^z_{\mathbf{k}\sigma}|\Downarrow_{\mathbf{k}\sigma}\rangle = 1
\end{aligned}$$

Thus, up to a phase

$$|\Uparrow_{\mathbf{k}\sigma}\rangle \sim \hat{c}^\dagger_{\mathbf{k}\sigma} |\Omega_{\mathbf{k}\sigma}\rangle \qquad |\Downarrow_{\mathbf{k}\sigma}\rangle \sim \hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma} |\Omega_{\mathbf{k}\sigma}\rangle$$

being $|\Omega_{\mathbf{k}\sigma}\rangle$ the $(\mathbf{k}, \sigma)$ Hilbert subspace vacuum.

It follows that composing Eqns. (A.24) and (A.25) the ground state $|\Psi\rangle$ can be written as:

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k}\in\mathrm{MBZ}} \left[\sin\theta_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}\sigma} + e^{2i\zeta_{\mathbf{k}\sigma}}\cos\theta_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right] |\Omega_{\mathbf{k}\sigma}\rangle \tag{A.26}$$

having we absorbed in the relative phase $e^{2i\zeta_{\mathbf{k}\sigma}}$ the two states phases differences as well as the original sign difference of Eq. (A.25). Now, inserting the explicit form of the ground state from equation above:

$$\begin{aligned}
\langle\overline{\Downarrow}_{\mathbf{k}\sigma}|\hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^\dagger_{\mathbf{k}\sigma}|\overline{\Downarrow}_{\mathbf{k}\sigma}\rangle &= \left(e^{2i\zeta_{\mathbf{k}\sigma}} + e^{-2i\zeta_{\mathbf{k}\sigma}}\right)\sin\theta_{\mathbf{k}\sigma}\cos\theta_{\mathbf{k}\sigma} \\
&= \cos 2\zeta_{\mathbf{k}\sigma}\sin 2\theta_{\mathbf{k}\sigma}
\end{aligned}$$

but from Eq. (A.11)

$$\begin{aligned}
\langle\overline{\Downarrow}_{\mathbf{k}\sigma}|\hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^\dagger_{\mathbf{k}\sigma}|\overline{\Downarrow}_{\mathbf{k}\sigma}\rangle &= \langle\overline{\Downarrow}_{\mathbf{k}\sigma}|\hat{s}^x_{\mathbf{k}\sigma}|\overline{\Downarrow}_{\mathbf{k}\sigma}\rangle \\
&= \sin 2\theta_{\mathbf{k}\sigma}
\end{aligned}$$

since the pseudofield has no y component. This implies necessarily

$$\cos 2\zeta_{\mathbf{k}\sigma} \quad \Longrightarrow \quad \zeta_{\mathbf{k}\sigma} = k\pi \quad \text{with} \quad k \in \mathbb{Z}$$

and lets us conclude from Eq. (A.26)

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k}\in\mathrm{MBZ}} \left[\sin\theta_{\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{k}\sigma} + \cos\theta_{\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right]|\Omega_{\mathbf{k}\sigma}\rangle \tag{A.27}$$

For low-temperature physics, we can expect the ground-state density matrix to be highly dominated by this BCS-like zero-temperature ground-state, which highlights the essential features of the system. Finally, using Eq. (A.23), we check that correctly the ground-state describes a perfect degenerate $\gamma^{(-)}$ quasiparticles Fermi gas,

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k}\in\mathrm{MBZ}} \gamma^{(-)\dagger}_{\mathbf{k}\sigma}|\Omega_{\mathbf{k}\sigma}\rangle \tag{A.28}$$

The AF phase can be thought of as a long-range coordination between fermionic states related by a $\boldsymbol{\pi}$ shift in reciprocal space, producing a new free gas where all interactions have been "condensed" in these AF pairs.

A.1.4 Spin degeneracy removal

Up to now, we wrote explicitly the σ index. However,

$$\epsilon_{\mathbf{k}\uparrow} = \epsilon_{\mathbf{k}\downarrow} \qquad \Delta_{\mathbf{k}\uparrow} = -\Delta_{\mathbf{k}\downarrow}$$

thus $E_{\mathbf{k}\uparrow} = E_{\mathbf{k}\downarrow}$. Since the gap function sign does has not a measurable impact, we might as well take the spin index as a pure degeneracy and from now on work in the $\sigma =\uparrow$ sector, dropping the σ index. We can further simplify by dropping all indices from the gap function,

$$\Delta_{\mathbf{k}\uparrow} = mU \equiv \Delta$$

being the latter purely s-wave. This is a property of the pure HM: for the EHM, the gap acquires a non-trivial topological structure.

A.1.5 Chemical potential and system density

Using the fact that

$$\left\langle \hat{\Gamma}^{\dagger}_{\mathbf{k}\sigma}\hat{\Gamma}_{\mathbf{k}\sigma}\right\rangle = \left\langle \hat{\Psi}^{\dagger}_{\mathbf{k}\sigma}\hat{\Psi}_{\mathbf{k}\sigma}\right\rangle$$

due to the $W_{\mathbf{k}\sigma}$ unitarity properties, the total number of electrons is given by

$$\begin{aligned} N &= \sum_{\mathbf{k}\sigma}\langle\hat{n}_{\mathbf{k}\sigma}\rangle \\ &= \sum_{\sigma}\sum_{\mathbf{k}\in\mathrm{MBZ}}\langle\hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle \\ &= \sum_{\sigma}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left\langle\hat{\Gamma}^{\dagger}_{\mathbf{k}\sigma}\hat{\Gamma}_{\mathbf{k}\sigma}\right\rangle \\ &= 2\sum_{\mathbf{k}\in\mathrm{MBZ}}\mathrm{Th}^{(+)}_{\mathbf{k}}[\beta,\mu] \end{aligned} \tag{A.29}$$

being f be the Fermi-Dirac distribution at inverse temperature β and chemical potential μ,

$$f(\epsilon;\beta,\mu) = \frac{1}{e^{\beta(\epsilon-\mu)}+1}$$

The 2 prefactor appeared because of spin degeneracy. Since the gapped bands refer to the smaller MBZ, thus exhibit the periodicity

$$E_{\mathbf{k}} = E_{\mathbf{k}+\boldsymbol{\pi}}$$

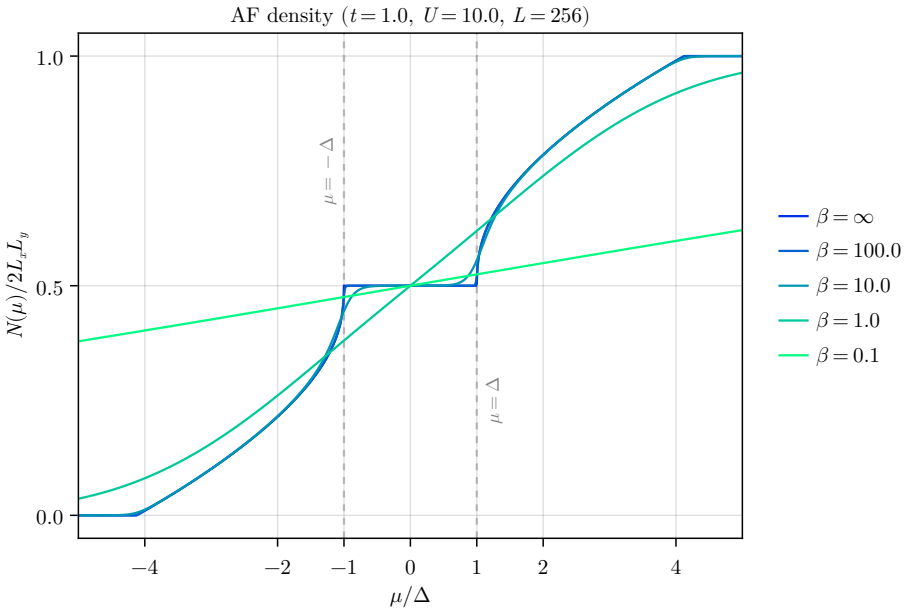


Figure A.5 | Electronic density in the AF phase as a function of the chemical potential, measured in relation to the gap $\Delta = mU$, at various temperatures.

it holds equivalently

$$N = \sum_{\mathbf{k}\in\mathrm{BZ}} \mathrm{Th}^{(+)}_{\mathbf{k}}[\beta, \mu] \tag{A.30}$$

as is obvious. The 2 prefactor was absorbed in doubling the integration region, MBZ $\to$ BZ. Next sections are devoted to simplify the above theoretical results in order to obtain an algorithmic estimation for the magnetization m, which is just the AF instability order parameter.

In Fig. A.5 the density $n(\mu)$ is plotted as a function of μ. As is evident, at nearly zero temperature ($\beta \to \infty$) the system is half filled for $-\Delta < \mu < \Delta$. In order to obtain a finite doping, it must be $|\mu| > \Delta$; which means, the chemical potential needs to cross the electronic bands. This HF solution leads invariably to a rather unusual metallic AF at any finite doping.

A.1.6 Self-consistency gap equation and magnetization

A convergence algorithm can be designed to find the Hartree-Fock solution to the model. Ultimately, we aim to extract m self-consistently. Combining Eqns. (A.5), (A.6) and the definition of Eq. (A.8), we get

$$\frac{1}{2L_xL_y}\sum_{\mathbf{r}}(-1)^{x+y}\hat{n}_{\mathbf{r}\sigma} = \sum_{\mathbf{k}\in\mathrm{MBZ}}\left[\hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^{\dagger}_{\mathbf{k}\sigma}\right]$$

thus

$$\begin{aligned} m &= \frac{1}{2L_xL_y}\sum_{\mathbf{r}}(-1)^{x+y}\langle\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}\rangle \\ &= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left\langle\left(\hat{\psi}_{\mathbf{k}\uparrow} + \hat{\psi}^{\dagger}_{\mathbf{k}\uparrow}\right) - \left(\hat{\psi}_{\mathbf{k}\downarrow} + \hat{\psi}^{\dagger}_{\mathbf{k}\downarrow}\right)\right\rangle \end{aligned}$$

Using now Eq. (A.11) we get the self-consistency equation for the HFP m,

$$\begin{aligned} m &= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left[\langle\hat{s}^x_{\mathbf{k}\uparrow}\rangle - \langle\hat{s}^x_{\mathbf{k}\downarrow}\rangle\right] \\ &= -\frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\mathrm{Re}\{\Delta_{\mathbf{k}\uparrow}\}-\mathrm{Re}\{\Delta_{\mathbf{k}\downarrow}\}}{2E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta,\mu] \\ &= -\frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\Delta}{2E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta,\mu] \end{aligned} \tag{A.31}$$

The magnetization m is the physical order parameter of the phase transition from the normal phase to the antiferromagnetic one. Within such phase transition, in fact, it is the expectation value

$$\langle\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle$$

that goes from zero (normal phase, free Fermi gas) to a non-zero value (AF). As is evident from equations above, m is just its (only) s-wave component.

Consider specifically the half-filled situation and multiply Eq. (**??**) on both sides by U. Since due to symmetry $\mu = 0$, we have the finite-temperature self-consistent equation

$$\Delta = \frac{U}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\Delta}{\sqrt{\epsilon^2_{\mathbf{k}}+\Delta^2}}\tanh\left(\frac{\beta}{2}\sqrt{\epsilon^2_{\mathbf{k}}+\Delta^2}\right)$$

where we used the relation

$$\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta,0] = -\tanh\left(\frac{\beta E_{\mathbf{k}}}{2}\right)$$

In thermodynamic limit $L\to\infty$

$$\frac{1}{U} \simeq \frac{1}{2}\int_{\mathrm{MBZ}}\frac{\mathrm{d}\mathbf{k}}{(2\pi)^2}\frac{1}{\sqrt{\epsilon^2_{\mathbf{k}}+\Delta^2}}\tanh\left(\frac{\beta}{2}\sqrt{\epsilon^2_{\mathbf{k}}+\Delta^2}\right)$$

Substitute the momentum integral with an integral over energies, in the low temperature limit, $\beta\to\infty$ (thus approximate the hyperbolic tangent with 1)

$$\frac{1}{U} \simeq \frac{1}{2}\int_{\mathrm{bands}}\frac{\mathrm{d}\epsilon\rho(\epsilon)}{\sqrt{\epsilon^2+\Delta^2}}$$

[To be continued...]

A.1.7 Instability of the commensurate AF solution and atypical metallic bands

A key point to be discussed is the known instability of the AF HF solution of the EHM. As is thoroughly discussed in [**singh1990collective**], the solution we presented above is only stable at half-filling, becoming hardly unstable at infinitesimal doping. This is due to the commensurate nature of the established AF phase: we impose SSB to take place at wavevector $\boldsymbol{\pi}$, shrinking the BZ to exactly half its size and giving back on the newly defined MBZ two bands $\pm E_{\mathbf{k}}$. This procedure actually works well for the half-filled case: the magnetization essentially depends on the scattering rate between points on the FS connected by a $\boldsymbol{\pi}$ vector. The half filled case, taking advantage of the natural nesting property of the bare bands,

$$\forall\mathbf{k}\in\partial\mathrm{MBZ}:\ \epsilon_{\mathbf{k}} = \epsilon_{\mathbf{k}+\boldsymbol{\pi}}$$

(being ∂MBZ the boundary of the MBZ) exhibits a magnetization at infinitesimal interaction. For a many body state resulting from continuous SSB, to establish a given phase means to show the presence of a stable gapless Goldstone mode associated to the broken symmetry. Such a Goldstone mode is essentially a sharp peak at zero frequency for some dressed response function. For the

AF phase, it follows from the reduced $\mathrm{U}^z(1)$ rotational symmetry and is related to the proper transverse pseudospin-pseudospin response function, 2×2 matrix equation,

$$\tilde{\chi}_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k},\omega) = \frac{\chi_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k},\omega)}{1 - U\chi_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k},\omega)} \tag{A.32}$$

being χ the free transverse response. The pseudospin hereby considered is the one of Eq. (A.10). As is shown in [**singh1990collective**], for the commensurate AF SSB at wavevector $\boldsymbol{\pi}$ the condition for the presence of a stable gapless mode reduces to a 2×2 matrix equation,

$$\text{Stability condition:} \qquad \mathbb{I}_{2\times 2} - U\left[\chi_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\boldsymbol{\pi},0)\right] \geq 0 \tag{A.33}$$

with $\left[\chi_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\boldsymbol{\pi},\omega)\right]$ the response function matrix obtained by considering a matrix response function for a two-sites unit cell. The above equation is to be interpreted as a matricial equation [Explain better..]. Considering the eigenvalue of largest amplitude of the matrix λ_δ at filling δ, the one most prone to instabilities, it is found that

$$\begin{aligned}
\lambda_0(\mathbf{k} \to \boldsymbol{\pi}, 0) &= \frac{1}{U} && \text{(half filling)} \\
\lambda_\delta(\mathbf{k} \to \boldsymbol{\pi}, 0) &= \frac{1}{U} + \frac{2\epsilon_{\mathrm{F}}}{\Delta} N(\epsilon_{\mathrm{F}}) && \text{(finite doping)}
\end{aligned}$$

The reason for the distinction above is the intrinsic metallic nature of the system for infinitesimal doping. At half-filling the system is an insulator, therefore no Fermi energy is defined, as well as no FS. Evidently this implies

$$\begin{aligned}
1 - U\lambda_0(\mathbf{k} \to \boldsymbol{\pi}, 0) = 0 \quad &\text{(half filling)} \quad \Longrightarrow \quad \text{stable Goldstone mode} \\
1 - U\lambda_\delta(\mathbf{k} \to \boldsymbol{\pi}, 0) < 0 \quad &\text{(finite doping)} \quad \Longrightarrow \quad \text{unstable Goldstone mode}
\end{aligned}$$

For any finite doping, the commensurate AF phase becomes unstable and is destroyed by transverse spin fluctuations. Nevertheless, for the sake of completeness and because we noticed too late, we will study the metallic AF also at finite doping.

A.1.8 Hartree-Fock algorithm results in reciprocal space

The above sections offer a self-consistency equation for the only HFP m; W is determined by $\Delta = mU$ indeed. In an iterative HF fashion, Eq. (A.31) reads:

$$m_{i+1} = \frac{U}{2L_xL_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{m_i}{2E_{\mathbf{k}}[\Delta_i]} \left[f(-E_{\mathbf{k}}[m_i];\beta,\mu) - f(E_{\mathbf{k}}[m_i];\beta,\mu)\right] \tag{A.34}$$

Then a self-consistent algorithm to search for a self consistent estimate of the magnetization may be sketched, following the general idea of Sec. **??**:

1. Algorithm setup: initialize a counter $i = 1$ and choose:
   - the local repulsion U/t or equivalently the hopping amplitude t/U;
   - the coarse-graining of the BZ, which is, fix L_x and L_y. Then

     $$k_x = 2\pi n_x/L_x \qquad k_y = 2\pi n_y/L_y$$

     where $-L_j/2 \leq n_j \leq L_j/2$, $n_j \in \mathbb{Z}$;
   - the density n or equivalently the doping $\delta \equiv n - 0.5$;
   - the inverse temperature β;
   - the maximum number of iterations $p \in \mathbb{N}$;
   - the mixing parameter $g \in [0,1]$;
   - the tolerance parameters $\delta_m, \delta_n \in \mathbb{R}$ (respectively for the HFP m and density);

2. Select a random starting value $m_i > 0$ for $i = 1$;
3. Find the optimal chemical potential μ as follows:
   a) Compute $\pm E_{\mathbf{k}}[\Delta_i, U]$;
   b) Define the function of the chemical potential μ,
   $$\delta n(\mu; \beta, m_i, U) \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \left[f\left(-E_{\mathbf{k}}[U, m_i]; \beta, \mu\right) + f\left(E_{\mathbf{k}}[U, m_i]; \beta, \mu\right)\right] - n$$
   c) Find the root of δn,
   $$\delta n(\mu_i) = 0$$
   Then μ_i is the chemical potential which, at step i with HFP m_i, realizes the best approximation of density n;
4. Compute m_{i+1} using Eq. (A.34) with chemical potential μ_i and update the counter, $i \to i+1$;
5. Check if $|m_{i+1} - m_i| \leq \delta_m$:
   - If yes, halt;
   - If not: check if $i > p$
     - If yes, halt and consider choosing better tolerance and model parameters;
     - If not, define
     $$m_{i+1} \colon g m_{i+1} + (1-g) m_i$$
     (logical assignment notation used) and repeat from step 2.

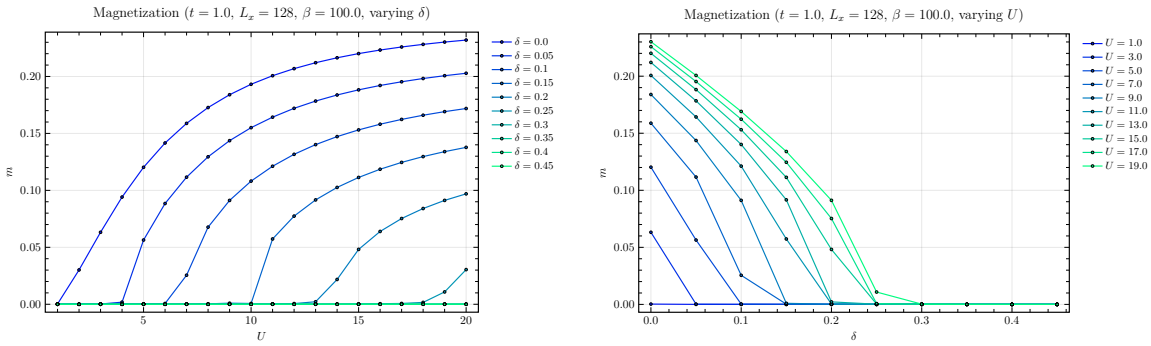
The algorithm sketched hereby was ran with the following setup:

```
UU = [U for U in 0.5:0.5:20.0]        # Local repulsions
LL = [256]                            # Lattice sizes
dd = [d for d in 0.0:0.05:0.45]       # Dopings
bb = [100.0, Inf]                     # Inverse temperatures
p = 100                               # Max number of iterations
dm = 1e-4                             # Tolerance on magnetization
dn = 1e-2                             # Tolerance on density
g = 0.5                               # Mixing parameter
```

In Fig. A.6 plots at $\beta = 100$ of the magnetization m is reported. As is evident in Fig. A.6a, doping disrupts AF ordering; the horizontal asymptote lowers as δ gets bigger, a mere consequence of the fact that the free space at each single-particle state in k-space shrinks as density increases; also, for $\delta > 0$, a finite magnetization m exists only for $U \geq U_c[\delta]$ (a critical value parametrically dependent on doping). Fig. A.6b shows the dependence of magnetization on doping at various fixed local repulsions U: once again, to dope the material disrupt AF ordering. Identical analysis have been performed for different temperatures, leading to analogous results as long as $\beta \gtrsim 10$ and to $m \simeq 0$ for higher temperatures (predictably).

A.2 (Impossibility of a stable) superconducting phase

Let us now move to the discussion of superconductivity in the HM. As for the AF phase, we use the results and analyses of this section as groundwork for the more sophisticated case of SC phase in the EHM. We anticipate here that BCS-like superconductivity cannot develop in the pure HM, due to the inherently repulsive nature of $\hat{H}_U$. BCS theory admits, for an half filled model, finite Cooper fluctuations for an infinitesimal attractive interaction. However, as will be discussed, by the means of MFT it is perfectly possible and coherent to derive a self-consistent superconducting solution to the HM. The point here is that such self-consistent solution sets to zero Cooper fluctuations.



a | Local repulsion U/t depencence.

b | Doping $\delta = n - 0.5$ dependence.

Figure A.6 | Plots of the zero-temperature AF instability order parameter m as a function of both the local repulsion U/t (Fig. A.6a) and the doping $\delta = n - 1/2$ (Fig. A.6b).

As done for the AF phase in Sec. A.1 and following the introductory discussion of Sec. [Missing], let us start from the discussion of the phase symmetries. We aim to describe a uniform-density superconductor,

$$\langle \hat{n}_{i\sigma} \rangle = n$$

taking into account Cooper RWCs,

$$\langle \hat{c}^\dagger_{i\sigma} \hat{c}^\dagger_{i\overline{\sigma}} \rangle$$

which, having we broken $U^{\mathrm{c}}(1)$ charge symmetry, we allow to fluctuate to non-zero values. For the pure HM, we know from Tab. **??** that both Hartree and Cooper pairing channels contribute to the RWCs set. We handle them separately.

A.2.1 MFT treatment of the Hartree channel

For the Hartree channel, knowing the deal with a uniform-density phase, we can re-use the derivation of Sec. A.1.1 simply by setting $m = 0$. We immediately get from Eq. (A.3):

$$\begin{aligned} \hat{H}_U &\simeq U \sum_{\mathbf{r}} \left[\hat{n}_{\mathbf{r}\uparrow} \langle \hat{n}_{\mathbf{r}\downarrow} \rangle + \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \hat{n}_{\mathbf{r}\downarrow} - \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \langle \hat{n}_{\mathbf{r}\downarrow} \rangle \right] + (\text{Cooper channel}) \\ &= nU \times \hat{N} - E^{(\mathrm{SC})}_{\mathrm{H}/U} + (\text{Cooper channel}) \end{aligned} \tag{A.35}$$

thus chemical potential renormalization $\tilde{\mu} \equiv \mu - nU$ is present also for the SC phase. The contraction energy comes immediately from Eq. (A.4),

$$E^{(\mathrm{SC})}_{\mathrm{H}/U} = L_x L_y \times n^2 U \tag{A.36}$$

We now move to Cooper channel.

A.2.2 MFT treatment of the Cooper channel

Recall the result of Eq. (**??**) which gave the Fourier-transformed version of the local repulsion, and perform a Cooper class Wick's contraction on it,

$$\begin{aligned} \hat{H}_U \simeq (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \Big[&\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \\ &+ \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \rangle - \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \rangle \langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \rangle \Big] \end{aligned} \tag{A.37}$$

We are not breaking translational invariance, thus only Cooper fluctuations with net zero total momentum are allowed. This means only $\mathbf{K} = \mathbf{0}$ contributes. Define as in Eq. (**??**) the pairing operator

$$\hat{\phi}_{\mathbf{k}\uparrow} \equiv \hat{c}_{-\mathbf{k}\downarrow} \hat{c}_{\mathbf{k}\uparrow} \tag{A.38}$$

and define $w_{\mathbf{q}}$ as in Eq. (2.4),

$$w_{\mathbf{q}} \equiv \langle \phi^\dagger_{\mathbf{q}} \rangle$$

Its s-wave part is given by:

$$w^{(s)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{q}\in \mathrm{BZ}} w_{\mathbf{q}} \tag{A.39}$$

Let $\hat{n}_{\mathbf{k}\sigma}$ as in Eq. (A.7). Eq. (A.35) reduces to

$$\hat{H}_U \simeq (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \left[\langle \hat{\phi}^\dagger_{\mathbf{q}} \rangle \hat{\phi}_{\mathbf{q}'} + \hat{\phi}^\dagger_{\mathbf{q}} \langle \hat{\phi}_{\mathbf{q}'} \rangle \right] - E^{(\mathrm{SC})}_{\mathrm{C}/U} \tag{A.40}$$

being $E^{(\mathrm{SC})}_{\mathrm{C}/U}$ the HM Cooper contraction energy,

$$\begin{aligned} E^{(\mathrm{SC})}_{\mathrm{C}/U} &\equiv \frac{U}{L_x L_y} \sum_{\mathbf{q}\mathbf{q}'} \langle \hat{\phi}^\dagger_{\mathbf{q}} \rangle \langle \hat{\phi}_{\mathbf{q}'} \rangle \\ &= L_x L_y \times U \left| w^{(s)} \right|^2 \end{aligned} \tag{A.41}$$

Let now

$$\Delta^{(s)} \equiv -U w^{(s)}$$

We finally reduce $\hat{H}_U$ to

$$\hat{H}_U \simeq (\text{Hartree channel}) - \sum_{\mathbf{q}\in \mathrm{BZ}} \left[\Delta^{(s)} \hat{\phi}_{\mathbf{q}} + \Delta^{(s)*} \hat{\phi}^\dagger_{\mathbf{q}} \right] - E^{(\mathrm{SC})}_{\mathrm{C}/U} \tag{A.42}$$

which can be easily mapped on a pseudospin model.

A.2.3 Nambu representation of the SC phase and diagonalization

Composing Eqns. (A.35) and (A.42), we get the full MFT hamiltonian

$$\begin{aligned} \hat{H} \simeq \sum_{\mathbf{k}\in \mathrm{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}\sigma} \left[\hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} - \hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \right] \\ - \sum_{\mathbf{q}\in \mathrm{BZ}} \left[\Delta^{(s)} \hat{\phi}_{\mathbf{q}} + \Delta^{(s)*} \hat{\phi}^\dagger_{\mathbf{q}} \right] + nU \times \hat{N} - \left[E^{(\mathrm{SC})}_{\mathrm{H}/U} + E^{(\mathrm{SC})}_{\mathrm{C}/U} \right] \end{aligned} \tag{A.43}$$

We now proceed precisely as in Sec. A.1.2. Define the SC Nambu spinor

$$\hat{\Phi}_{\mathbf{k}} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\uparrow} \\ \hat{c}^\dagger_{-\mathbf{k}\downarrow} \end{bmatrix}$$

Define the pseudospin operator:

$$\hat{s}^\alpha_{\mathbf{k}} \equiv \hat{\Phi}^\dagger_{\mathbf{k}} \tau^\alpha \hat{\Phi}_{\mathbf{k}}$$

and the shifted bare bands,

$$\xi_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} - \tilde{\mu}$$

Notice we are using the chemical potential RMP. We employ from the start the spin-independency of the bare bands, as well as their pure s^*-wave structure, since no Fock interaction in the present context chan change that. The following chain holds:

$$\begin{aligned} \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{-\mathbf{k}\downarrow} &= \hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}_{\mathbf{k}\uparrow} + \hat{c}^\dagger_{-\mathbf{k}\downarrow} \hat{c}_{-\mathbf{k}\downarrow} \\ &= \hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{-\mathbf{k}\downarrow} \hat{c}^\dagger_{-\mathbf{k}\downarrow} + \hat{\mathbb{I}}_{\mathbf{k}} \\ &= \hat{\Phi}^\dagger_{\mathbf{k}} \tau^z \hat{\Phi}_{\mathbf{k}} + \hat{\mathbb{I}}_{\mathbf{k}} \\ &= \hat{s}^z_{\mathbf{k}} + \hat{\mathbb{I}}_{\mathbf{k}} \end{aligned} \tag{A.44}$$

Then, considering the kinetic part of the gran-canonical hamiltonian,

$$\begin{aligned}\hat{H}_t - \tilde{\mu}\hat{N} &= \sum_{\sigma}\sum_{\mathbf{k}\in\mathrm{BZ}} (\epsilon_{\mathbf{k}\sigma} - \tilde{\mu})\,\hat{n}_{\mathbf{k}\sigma} \\ &= \sum_{\mathbf{k}\in\mathrm{BZ}} [\xi_{\mathbf{k}}\hat{n}_{\mathbf{k}\uparrow} + \xi_{-\mathbf{k}}\hat{n}_{-\mathbf{k}\downarrow}] \\ &= \sum_{\mathbf{k}\in\mathrm{BZ}} \xi_{\mathbf{k}}\hat{s}^z_{\mathbf{k}} - E^{(\mathrm{SC})}_{\mathrm{A}/t}\end{aligned}$$

being $E^{(\mathrm{SC})}_{\mathrm{A}/t}$ the "anticommutation energy" coming from the kinetic term,

$$E^{(\mathrm{SC})}_{\mathrm{A}/t} \equiv -\sum_{\mathbf{k}\in\mathrm{BZ}} \xi_{\mathbf{k}} \tag{A.45}$$

The scalar term in the above equation is essential for estimating correctly the phase ground state free energy density. The full MFT gran-canonical hamiltonian in Nambu representation is obtained including Eq. (A.39),

$$\hat{H} - \mu\hat{N} \simeq \sum_{\mathbf{k}\in\mathrm{BZ}}\sum_{\sigma} \hat{\Phi}^\dagger_{\mathbf{k}} h_{\mathbf{k}} \hat{\Phi}_{\mathbf{k}} - \left[E^{(\mathrm{SC})}_{\mathrm{A}/t} + E^{(\mathrm{SC})}_{\mathrm{H}/U} + E^{(\mathrm{SC})}_{\mathrm{C}/U}\right] \tag{A.46}$$

being

$$h_{\mathbf{k}} \equiv \begin{bmatrix} \xi_{\mathbf{k}} & -\Delta_{\mathbf{k}} \\ -\Delta^*_{\mathbf{k}} & -\xi_{\mathbf{k}} \end{bmatrix}$$

while the off-diagonal gap is a constant,

$$\Delta_{\mathbf{k}} \equiv \Delta^{(s)}$$

This hamiltonian behaves as an ensemble of spins in local magnetic pseudofields

$$\mathbf{b}_{\mathbf{k}} \equiv \begin{bmatrix} -\operatorname{Re}\{\Delta_{\mathbf{k}}\} \\ -\operatorname{Im}\{\Delta_{\mathbf{k}}\} \\ \xi_{\mathbf{k}} \end{bmatrix} \qquad \tan 2\theta_{\mathbf{k}} \equiv \frac{\xi_{\mathbf{k}}}{\sqrt{\xi^2_{\mathbf{k}} + |\Delta_{\mathbf{k}}|^2}} \qquad \tan 2\zeta_{\mathbf{k}} \equiv \frac{\operatorname{Im}\{\Delta_{\mathbf{k}}\}}{\operatorname{Re}\{\Delta_{\mathbf{k}}\}} \tag{A.47}$$

precisely as was for the AF phase, with the essential difference that now the chemical potential RMP lies inside the z component of the pseudofield itself. This is a consequence of the Nambu spinors definition. We have the spin hamiltonian

$$\hat{H} - \mu\hat{N} \simeq \sum_{\mathbf{k}\in\mathrm{BZ}} \mathbf{b}_{\mathbf{k}} \cdot \hat{\mathbf{s}}_{\mathbf{k}} - \left[E^{(\mathrm{SC})}_{\mathrm{A}/t} + E^{(\mathrm{SC})}_{\mathrm{H}/U} + E^{(\mathrm{SC})}_{\mathrm{C}/U}\right] \quad \text{where} \quad \hat{\mathbf{s}}_{\mathbf{k}} = \begin{bmatrix} \hat{s}^x_{\mathbf{k}} \\ \hat{s}^y_{\mathbf{k}} \\ \hat{s}^z_{\mathbf{k}} \end{bmatrix} \tag{A.48}$$

As was for the AF phase, to diagonalize the hamiltonian means essentially to align the spin z axis with the direction of the pseudofield. Then, we diagonalize the hamiltonian via a set of rotations similarly to Eq. (A.15). The main difference is that in general the pseudofield can have a y component, thus the diagonalizing matrix is made of a sequence of two rotations, both counter-clockwise, the first around the z axis by an angle $2\zeta_{\mathbf{k}}$ to place the pseudofield in the xz plane, the second around the new y axis by an angle $2\theta_{\mathbf{k}}$ to align the z axis with the pseudofield,

$$\begin{aligned} W_{\mathbf{k}} &\equiv e^{-i\theta_{\mathbf{k}}\tau^y} e^{-i\zeta_{\mathbf{k}}\tau^z} \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}} & -\sin\theta_{\mathbf{k}} \\ \sin\theta_{\mathbf{k}} & \cos\theta_{\mathbf{k}} \end{bmatrix} \begin{bmatrix} e^{-i\zeta_{\mathbf{k}}} & \\ & e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & -\sin\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \\ \sin\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & \cos\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \end{aligned} \tag{A.49}$$

obtaining the diagonal matrix

$$d_{\mathbf{k}} \equiv W_{\mathbf{k}} h_{\mathbf{k}} W^\dagger_{\mathbf{k}} \quad \text{where} \quad d_{\mathbf{k}} \equiv \begin{bmatrix} E_{\mathbf{k}} & \\ & -E_{\mathbf{k}} \end{bmatrix}$$

and the bands

$$E_{\mathbf{k}} \equiv \sqrt{\xi_{\mathbf{k}}^2 + |\Delta_{\mathbf{k}}|^2}$$

Finally, the Nambu spinor for Bogoliubov quasiparticles is given similarly to Eq. (A.22)

$$\hat{\Gamma}_{\mathbf{k}} = W_{\mathbf{k}} \hat{\Phi}_{\mathbf{k}} \tag{A.50}$$

Explicitly,

$$\begin{aligned} \hat{\Gamma}_{\mathbf{k}} &= \begin{bmatrix} \cos\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & -\sin\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \\ \sin\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & \cos\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \begin{bmatrix} \hat{c}_{\mathbf{k}\uparrow} \\ \hat{c}^\dagger_{-\mathbf{k}\downarrow} \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} \hat{c}_{\mathbf{k}\uparrow} - \sin\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \\ \sin\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} \hat{c}_{\mathbf{k}\uparrow} + \cos\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \end{bmatrix} \equiv \begin{bmatrix} \hat{\gamma}^{(+)}_{\mathbf{k}} \\ \hat{\gamma}^{(-)}_{\mathbf{k}} \end{bmatrix} \end{aligned} \tag{A.51}$$

These operators act as free fermion fields on the Bogoliubov bands $\pm E_{\mathbf{k}}$.

A.2.4 Explicit zero-temperature SC ground state at half-filling

Reasoning precisely as in Sec. A.1.3, we can extract the half-filling zero temperature ground state, commonly referred to as the "BCS state". Eqns. (A.24) and (A.25) hold equivalently (ignoring the σ index and extending momentum tensor product to the entire BZ),

$$|\Psi\rangle = \bigotimes_{\mathbf{k}\in\mathrm{BZ}} \left[\cos\theta_{\mathbf{k}} \left|\Downarrow_{\mathbf{k}}\right\rangle - \sin\theta_{\mathbf{k}} \left|\Uparrow_{\mathbf{k}}\right\rangle\right]$$

Using now Eq. (A.44), we have

$$\begin{aligned} \langle\Downarrow_{\mathbf{k}}|\hat{s}^z_{\mathbf{k}} + \hat{\mathbb{I}}_{\mathbf{k}}|\Downarrow_{\mathbf{k}}\rangle &= 0 \\ \langle\Uparrow_{\mathbf{k}}|\hat{s}^z_{\mathbf{k}} + \hat{\mathbb{I}}_{\mathbf{k}}|\Uparrow_{\mathbf{k}}\rangle &= 2 \end{aligned}$$

so up to a phase

$$\left|\Downarrow_{\mathbf{k}}\right\rangle \sim \left|\Omega_{\mathbf{k}}\right\rangle \qquad \left|\Uparrow_{\mathbf{k}}\right\rangle \sim \hat{\phi}^\dagger_{\mathbf{k}} \left|\Omega_{\mathbf{k}}\right\rangle$$

Hence, we have

$$|\Psi\rangle = \bigotimes_{\mathbf{k}\in\mathrm{BZ}} \left[\cos\theta_{\mathbf{k}} + e^{2i\zeta_{\mathbf{k}}} \sin\theta_{\mathbf{k}} \hat{\phi}_{\mathbf{k}}\right] \left|\Omega_{\mathbf{k}}\right\rangle$$

The argument $2\zeta_{\mathbf{k}}$ of the phase factor is precisely the phase of the gap function, as can be shown easily.

A.2.5 Superconducting fluctuations suppression in the conventional HM

Let us report the SC analogous of the AF Eqns. (A.11) and (A.12):

$$\hat{s}^x_{\mathbf{k}} = \hat{\phi}_{\mathbf{k}} + \hat{\phi}^\dagger_{\mathbf{k}} \tag{A.52}$$

$$\hat{s}^y_{\mathbf{k}} = -i\left(\hat{\phi}_{\mathbf{k}} - \hat{\phi}^\dagger_{\mathbf{k}}\right) \tag{A.53}$$

while the analogous of Eq. (A.13) was given in Eq. (A.44). Let also the thermal factor

$$\mathrm{Th}^{(\pm)}_{\mathbf{k}}[\beta] \equiv f\left(E_{\mathbf{k}};\beta,0\right) \pm f\left(-E_{\mathbf{k}};\beta,0\right) \tag{A.54}$$

and the EVs:

$$\langle\hat{s}^x_{\mathbf{k}}\rangle = -\frac{\mathrm{Re}\{\Delta_{\mathbf{k}}\}}{2E_{\mathbf{k}}} \mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \tag{A.55}$$

$$\langle\hat{s}^y_{\mathbf{k}}\rangle = -\frac{\mathrm{Im}\{\Delta_{\mathbf{k}}\}}{2E_{\mathbf{k}}} \mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \tag{A.56}$$

$$\langle\hat{s}^z_{\mathbf{k}}\rangle = \frac{\xi_{\mathbf{k}}}{2E_{\mathbf{k}}} \mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \tag{A.57}$$

Up to now, the derivation was general. However, it is immediate to show that for the pure HM the SC phase can never establish. Recall Eq. (A.41): explicitly

$$\begin{aligned} w^{(s)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} \left\langle \hat{\phi}^\dagger_{\mathbf{k}} \right\rangle \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} \left\langle \hat{\Phi}^\dagger_{\mathbf{k}} \tau^+ \hat{\Phi}_{\mathbf{k}} \right\rangle \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} \left[\langle \hat{s}^x_{\mathbf{k}} \rangle + i \langle \hat{s}^y_{\mathbf{k}} \rangle \right] \\ &= -\frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} \frac{\Delta_{\mathbf{k}}}{E_{\mathbf{k}}} \mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \end{aligned} \tag{A.58}$$

Now: $\Delta_{\mathbf{k}} = \Delta^{(s)} = -U w^{(s)}$, thus if we define

$$U_{\mathrm{c}}[\beta] \equiv \left[\frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} \frac{\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta]}{E_{\mathbf{k}}} \right]^{-1}$$

(the reason for this notation will be clear in Sec. [Missing]) we end with the equation:

$$w^{(s)} = -\frac{U}{U_{\mathrm{c}}[\beta]} w^{(s)}$$

Since evidently $U_{\mathrm{c}}[\beta] > 0$ by definition, the above equation can be solved in $\mathbb{C}$ only by $w^{(s)} = 0$. This implication terminates the argument: superconductivity cannot establish self-consistently at this level in the HM.